The Vault rig

A computer that works only while its owner's own voice and hands are producing what it receives, and every agent action traces back to them. Six pieces, five checks, all required, day or night.

Prepared by Pablo Peniche **Status** Draft for discussion · not for distribution **Sources** 49, numbered inline

Also the command chain · hardware spec · full research report

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IN ONE PARAGRAPH

Six pieces, five checks, all required.

Every few seconds, Vault proves that the owner's own body is producing what the computer receives, and that every agent action traces back to it. **Earbuds** carry the voice and the ear. **Muscle-sensing wristbands and a signed keyboard** carry the hands. **An IR camera** carries the face. **The hub** ties them together and holds the keys. **A master agent and a checking gateway** carry every command down to the tools. All five checks must hold, in daylight or in the dark. Vault makes its own hardware: the earbuds, the wristbands, the keyboard, the camera module and the hub. Off-the-shelf products serve only as references and early prototypes.

Numbers are lab values from the cited studies, and every threshold has to be calibrated per owner. Nothing here has been tested end to end. "Proposal" marks design choices made for Vault rather than measured results.

01 · THE RIG

Six pieces, all made by Vault

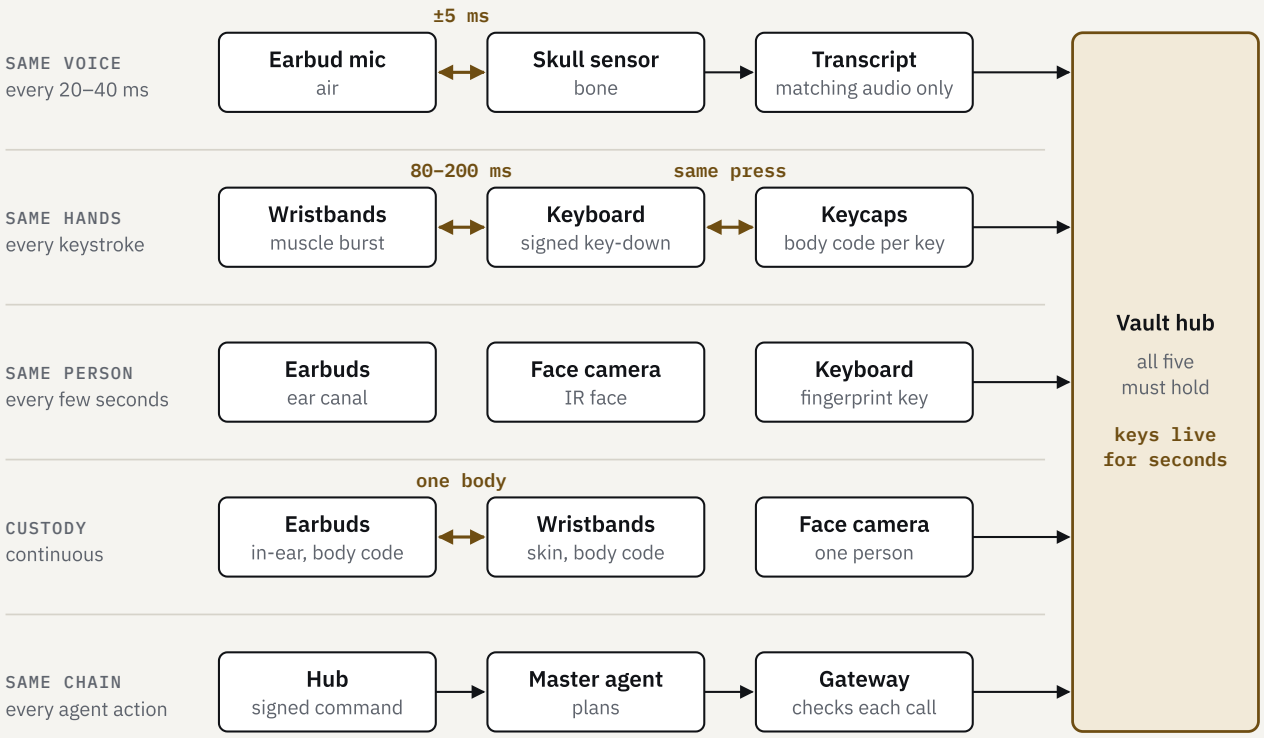
PIECE	WHERE	WHAT IT DOES	REFERENCE PARTS AND PROTOTYPES
Wired earbuds	Both ears	Ear-canal echo says *who*; a skull-vibration sensor and inner mic prove *your* voice for dictation; an in-ear sensor detects removal; a faint coded signal goes out through your skin to the wristbands	OpenEarable 2.0, an open research earbud, has the bone sensor and inner mics[1]; ear-canal ID reached 98.7% in research[2]; body-signal chips exist (Ixana Wi-R, vendor-claimed 5 Mb/s)[3]
Wristbands (sEMG)	Both wrists	Muscle signals that fire before every keystroke and show which hand and finger moved; a soft identity signal; skin contact for custody; they receive the earbuds' body code	Built to Meta's research spec: 16 channels at 2 kHz, dry electrodes[4]. Meta's own Neural Band lasts about 18 hours[5] but gives developers only gesture events[6]. Early prototypes can use MindRove: 8 channels at 500 Hz, $679–729[7]
Vault keyboard	Desk	Signs every key event with its own timestamp; electrode keycaps that hear the earbuds' body code on every press; a fingerprint key for identity, confirmations and a duress finger	Analog Hall-effect switches like the Wooting 80HE's[8]; a fingerprint key paired the way Apple's Touch ID keyboard pairs[9]; touch electrodes as in the DiamondTouch table[10]
Face camera	On the monitor	Infrared face on a signed path that software cannot feed. Its own infrared light works in the dark	A camera module that signs its frames, the protection Windows Enhanced Sign-in Security gives built-in cameras[11]
Vault hub	Desk, cabled to the computer	Runs the checks, keeps one clock, holds the keys, shows and reads confirmations, keeps the trace log	A secure element with a small screen and speaker, built like a hardware wallet
Master agent and gateway	Software, keys held in the hub	Turns verified commands into signed roots and checks every agent action	Vault software on open building blocks: narrowing tokens[12], injection-resistant planning[13]; see the command chain

The earbuds, keyboard and camera are cabled to the hub, because Bluetooth relays survive encryption and add only about 8 ms[14]. The wristbands pair with per-device keys and sign their own timestamps. The earbuds' cable also closes the body-signal loop through a shared ground. The bands pick the code up at the wrist all the time, and the keycaps pick it up on every press. That proves the earbuds, the wrists and each pressing finger belong to one body. It is the DiamondTouch table's design in reverse: in 2001 that table sent a signal through each user to a receiver in their chair, so it knew who touched where[10].

02 · FIVE CHECKS

Five checks, all required

Five checks run at once, and every piece feeds at least two of them, so faking or stealing one piece breaks another check.



How the pieces check each other. No piece is trusted on its own. Speech counts only when the skull sensor felt it[17]. A keystroke counts only when a wrist-muscle burst came first and its keycap heard the body code[23, 29]. Earbuds and wristbands prove they are on one body by passing a code through the skin[29]. An agent action counts only when it traces to a signed command[12, 13]. The hub keeps the computer's keys alive for a few seconds at a time, only while all five checks hold.

CHECK	QUESTION	HOW	STOPS	DOES NOT STOP
Same voice	Did the wearer's own throat produce this speech?	Each voiced 20–40 ms piece the earbud mic hears must also show up as skull vibration in the same earbud	Loudspeakers, AI voice clones, injected audio	The owner speaking under coercion

CHECK	QUESTION	HOW	STOPS	DOES NOT STOP
Same hands	Did the owner's own fingers press these keys?	Each signed key-down pairs with exactly one muscle burst from the right hand and a plausible finger, starting 80–200 ms before it, and its keycap heard the body code; every burst must have a key	AI agents, malware, injected USB devices, another person typing, a key pushed mechanically	A forged band signal; someone touching the owner while pressing a key
Same person	Is this the owner?	Ear canal in speech pauses, IR face, fingerprint key, and the owner's own muscle pattern	Another person wearing the gear	A coerced owner
Custody	Has anything changed since the owner last proved who they are?	Earbuds in, bands on the skin, the earbuds' body code reaching both wrists, one person in view	Gear handed over mid-session; earbuds and bands split between two people	Someone touching the owner, since the body code passes through touch
Same chain	Does every agent action trace to a verified command?	Hub-signed roots, tokens that only narrow, a gateway that checks every call	Prompt-injected or rogue agents acting on their own	A verified command that was itself a mistake

Each attack fails a different check. A voice clone fails *same voice*. An agent, a malicious USB cable or a robot pressing keys fails *same hands*. Stolen gear fails *same person*. A swap mid-session fails *custody*. An agent tricked by a web page fails *same chain*. Only a coerced owner passes all five, which is why the fingerprint key also takes a duress finger.

03 · DICTATION

The same-voice check

AirPods already ship a "speech-detecting accelerometer"[15]. Apple's patent describes it as an in-canal sensor sampled at 2–6 kHz, used for noise suppression and closed to developers[16]. Research turns the same signal into a security check. VAuth accepted 97% of genuine commands with 0.09% false positives across 18 users, and airborne sound could fool it only from within about 30 cm at 90 dB[17]. AirBone reached a 1.1% equal error rate for liveness and 4.0–4.5% against replay and impersonation, with 31 users[18].

The check runs on the earbud itself (proposal). The skull sensor and inner mic share the earbud's clock and are signed by it. The physical lag between the two paths is under 1 ms, since the skull carries sound across the head in about 0.4 ms[19]. Every 20–40 ms of voiced speech must line up within ±5 ms, and only matching audio reaches transcription. Words get ±50 ms of slack, because even the best aligner places only about 73% of word boundaries within 25 ms[20]. Dictated commands become signed roots only from this verified audio.



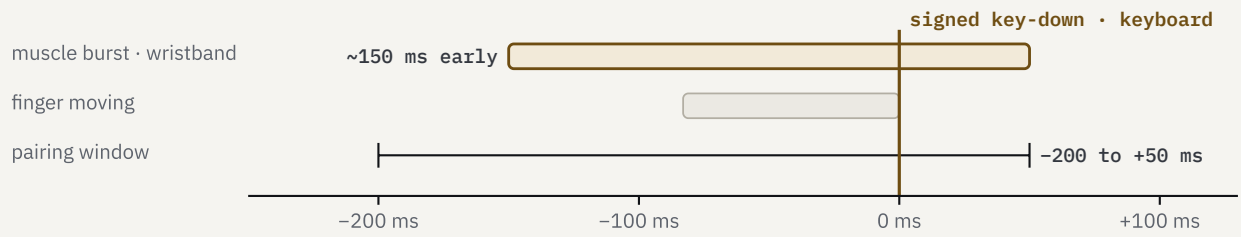
Speech counts only if your skull felt it. The earbud's microphone hears everything in the room. The skull sensor in the same earbud feels only the wearer's own voice. Each voiced piece must appear in both, on the earbud's own clock, within a few milliseconds, or it never reaches transcription[17, 19]. The skull signal shows *when* you spoke precisely, and *what* you said only roughly[21].

Three limits remain:

- **What vs when.** The skull signal proves *when* you spoke with high confidence, but *what* you said only roughly. The best ear-worn systems recognize about 95% of words from a 100-word vocabulary[21]. Unvoiced sounds such as *s*, *f* and *t* barely reach the skull sensor[16].
- **Sensor forging.** Loud sound at a sensor's resonant frequency biased 75% of 20 tested accelerometer models[22]. The earbud must sign its data and filter out-of-band vibration.
- **Untested attacks.** No one has tested modern AI voice clones against this check, or speech injected while the owner is talking[17]. Jaw movement also changes the ear canal's shape, so ear-canal ID runs in speech pauses[21].

Wristbands and keycaps

Forearm muscles fire about 150 ms before a key bottoms out and stop about 50 ms after it, and the finger starts moving about 83 ms before[23]. A detector can place a keystroke from wrist signals to within about 21 ms, and finds 94% of keystrokes within ±50 ms[23]. Vault pairs every signed key-down with exactly one burst that starts 80–200 ms before it, from the right hand and a plausible finger. Every burst must also have a key. Pairing both ways closes the hole that broke the earlier ZEBRA wristband scheme, where the attacker's own typing decided when comparisons ran[24, 25].



One keystroke, in milliseconds. The wrist muscles start firing about 150 ms before the key bottoms out, and the finger starts moving about 83 ms before[23]. Each signed key-down must pair with exactly one burst inside the window, and each burst with one key. An injected keystroke has no burst behind it.

Single keys are too noisy to judge. At 4.4 keys a second the bursts overlap, and even a decoder trained on its owner misreads 7–11% of characters[26]. So Vault decides over windows of 20–50 keys, about 5–12 seconds (proposal). That catches injected input within seconds. It cannot catch one swapped key typed by the same finger. Key travel and switch type shift the timing by 20–40 ms, so the template is calibrated per owner and per keyboard[27]. The bands also identify their wearer: wrist muscle patterns told people apart at about a 2.5% equal error rate across days, in a 43-person lab study[28]. Muscles work the same in the dark.

Meta's research band is the model to build to: 16 channels at 2 kHz with very low noise, decoding handwriting at about 21 words a minute[4]. Its consumer Neural Band runs about 18 hours per charge[5] but exposes only gesture events[6]. So Vault builds its own band to that spec, with signed firmware and its own timestamps. Today's raw-signal bands are a step down: MindRove has 8 channels at 500 Hz, costs $679–729, and lasts 4–6 hours per charge[7].

The **keycaps** add proof for each individual key. The earbuds send a faint coded signal at roughly 100 kHz–1 MHz that spreads over the skin[29]. The current is a few microamps, about 2,000 times below the contact current where pain begins[30, 31]. The bands hear the code constantly at the wrist. Each keycap carries an electrode that hears it during the press, so a key pushed by a mechanical plunger or by someone else's finger arrives without it. A keystroke lasts only about 100–150 ms, and earlier security prototypes at 0.5–1 kb/s needed touches over 300 ms to deliver a full 128-bit code[32]. So Vault's link must run at 10–100 kb/s or more, which body-network chips advertise[3]. The signal mostly stays on the body: an Oxford prototype told the body channel from fakes 99.3% of the time[33].

Two limits remain. Coupling starts about a centimetre before contact, so the code proves that *a finger of the owner's body* pressed the key, not which finger; the bands supply which finger. And anyone touching the owner carries the code too, while a live relay through a second person can be caught only by time or distance bounds, not by the code itself[34], so the face camera also requires exactly one person in view (proposal).

05 · COMMANDS AND AGENTS

Commands and agents

Everything the verified owner says or types becomes a hub-signed command. A master agent plans only from those commands. It reads web pages, email and files through a quarantined reader that has no tools, and it hands sub-agents tokens that can only narrow[12, 13]. A gateway outside any AI model checks every action against its chain. It asks the owner when an action is consequential or was shaped by untrusted data, and it logs everything, including refusals. The full design is on the command chain page.

06 · IF A PIECE IS MISSING

If a piece is missing

Missing data lets trust fade for a few seconds. Contradicting data locks at once: a different face, a key with no muscle burst or body code, speech with no skull vibration (proposal).

PIECE OFF OR FAILING	WHAT GOES BLIND	WHAT VAULT DOES (PROPOSAL)
Earbuds out	Voice, ear identity, body code	Typing and dictation stop reaching the computer at once. When the earbuds go back in, the fingerprint key and face re-anchor
A wristband off	Same hands for that hand, body code	That hand's keys stop reaching the computer at once. When it goes back on, the fingerprint key re-anchors
A key arrives without the body code	Same hands	Treated as a contradiction: the key is dropped and vault actions lock until the owner re-anchors
Wristband battery low	Nothing yet	Warns ahead of time. A second pair swaps in during a fingerprint touch
Face out of view	Face identity	Nothing for short gaps, while the ear check carries identity. Vault actions wait for the face
A second person in view	Custody	Vault actions pause until the owner confirms with the fingerprint key
Unknown keyboard or USB device	Signed input	Its input is blocked
Hub unplugged	Everything	Keys expire within seconds; the computer locks

07 · OPTIONS SCORED

The options, scored

OPTION	VERDICT	WHY
Wired earbuds	**Must-have**	Voice binding for dictation, all-day ear identity, and the source of the body code. The cable rules out radio relay
sEMG wristbands, both wrists	**Must-have**	The only signal that ties each keystroke to muscles, in any light[23]. Also a soft identity signal and the receiver for the body code
Vault keyboard: signed keys, electrode keycaps, fingerprint key	**Must-have**	Every key event signed at the switch and tied to a finger of the owner's body. Identity on demand, the confirm button and a duress finger
IR face camera on an attested path	**Must-have**	The owner's face whenever they face the screen, day or night, over a path software cannot feed[11]
Master agent and gateway	**Must-have**	Every agent action traced to a verified command
Hands camera	**Optional**	Infrared, so darkness is no problem[35]. But keys sit under 1 cm apart, and one tracker missed 7–10% of keystrokes[36]. A cross-check, not a requirement
Chest or microneedle patch	**Optional**	A single-use custody seal for the highest-stakes sessions[37]
Heartbeat timing check	**In reserve**	Needed only if Vault must accept ordinary webcams; see section 8
Ring, watch, Apple Watch ranging	**Drop**	Wrist pulse fails as identity in real use[38]. Watch ranging proves a token is near, not who is typing[39]
Nymi-style ECG band	**Drop**	Replayed ECG fooled it 81% of the time[40]
RGB webcam face	**Demote**	Never identity, because virtual cameras and injected frames can feed it[41, 42]
DNA or metabolome	**Drop**	Rapid DNA takes about 90 minutes[43]; metabolic profiles re-identify only about 40% of people after 7 years[44]

08 · IN RESERVE

Heartbeat timing and a hands camera

Heartbeat timing. After each heartbeat, the pulse reaches the ear about 120–200 ms later and the finger about 200–270 ms later[45, 46]. Successive beats differ by about 42 ms, so a forger can't predict them[47]. A live deepfake's face pulse comes from the attacker and arrives late, because rendering adds at least ~43 ms per frame[41, 48]. With a camera that signs its frames, this check is redundant.

Hands camera. A downward infrared tracker can confirm that a finger landed on each reported key. Headset cameras detected touches at an F1 score of 0.99 and named the finger at 0.96[49]. It is worth adding as a cross-check if the bands misfire in testing, but it cannot replace them.

What is not proven yet

- **None of the core checks has been measured.** That covers skull-versus-mic matching against modern voice clones, per-key pairing of muscle bursts with signed key events, and a body code carried from the ear to the wrists and keycaps. No one has published muscle-to-keystroke pairing as a security check, and every number above comes from a separate lab study.
- **Vault makes all of the hardware.** None of these exists as a product: wired earbuds with a skull sensor, inner mic and body-signal transmitter; a wristband with Meta-class signal quality, raw access, signed firmware and all-day battery; a keyboard with signed analog keys, electrode keycaps and a fingerprint key; a camera module that signs its frames; and the hub.
- **Forging the band's signal.** ECG replayed into a band's electrodes worked 81% of the time on the Nymi[40]. Whether muscle signals can be injected into a 16-channel band is untested. The band stream is also a keylogger, so raw signals never leave the hub.
- **Touch passes the body code.** A second person touching the owner carries it to the keycaps too, and a live relay needs time or distance bounds to catch it. The one-person rule carries that risk.
- **Daily wear.** Earbuds plus two bands. Meta's band shows about 18 hours per charge is possible[5]. No study of consumer earbuds worn for 8+ hours was found.
- **Coercion.** All five checks pass for a coerced owner. The duress finger, delays and second approvers handle that; see the full report.
- **The agent chain's open problems.** No planner design has been tested against adaptive attackers, and turning speech into permissions is itself a trust boundary; see the command chain.

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Vault · the rig · October 2026. A specific build drawn from the full research report, which covers the threat model, law, market, duress and succession. Earlier gate-based design: v1 (PDF).

The command chain

How every action an AI agent takes on a Vault computer traces back, through signed permissions that only narrow, to a command the verified owner actually spoke or typed.

Prepared by Pablo Peniche **Status** Draft for discussion · not for distribution **Sources** 21, numbered inline

Also the rig · hardware spec · full research report

IN ONE PARAGRAPH

Every agent action traces back to a verified human.

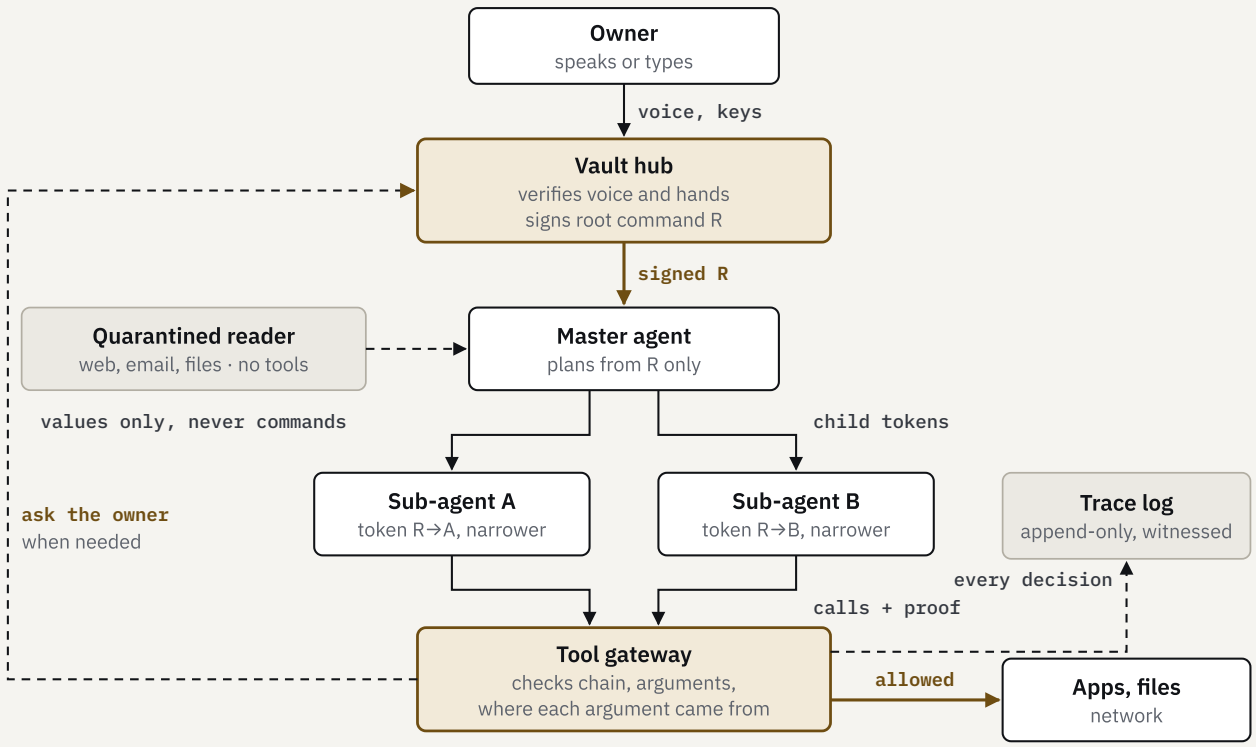
The Vault hub turns each verified human command into a signed root. A master agent plans from verified commands only and hands sub-agents permission tokens that can only narrow. A tool gateway, which is Vault-controlled code rather than an AI model, checks every action against its chain before it runs. It asks the owner when an action is consequential or was shaped by untrusted data, and it writes every allowed, refused and unexplained action to a tamper-evident log. Each piece exists today in some form. No standard and no evaluated system joins them to a continuously verified human[14, 19].

"Proposal" marks design choices made for Vault. Draft standards are labelled as drafts. This page covers the agent side; how Vault verifies the human is in the rig and the full report.

01 · THE CHAIN

From your voice to every tool call

The master agent is the only agent that receives commands, and it only receives commands the hub has signed. Everything below it acts on a narrower copy of that authority. The gateway at the bottom is the only path to tools, files and the network.



From your voice to every tool call (proposal). Only the hub can start a chain, and only after it has verified the voice or hands behind a command. The master agent plans from signed commands; untrusted content goes to a reader that has no tools[1]. Each hop carries a token that can only narrow[2, 3]. The gateway, not any agent, decides what runs. It logs every decision to a witnessed log[9, 11] and sends anything unusual back to you through the hub.

Five rules

01 **Only the hub can start a chain.** A command becomes a root only after the hub has verified the voice or hands that produced it. The root is the hub's signature over the exact transcript, plus the evidence it verified. For consequential commands, the hub reads back the scope the master agent derived ("send Ana the Q3 PDF") before signing, because turning words into permissions is itself a trust boundary[5].

02 **Authority only narrows.** Each sub-agent gets a child token that hashes its parent, is bound to the sub-agent's own key, and can only shrink the allowed tools, arguments, depth and lifetime. Biscuit tokens ship today[3]; UCAN is at release candidate[4]; an IETF individual draft (June 2026) specifies this for agent chains exactly, and leaves the link from the root key to a human as "a deployment responsibility"[2]. That missing link is what the hub provides. A sub-agent cannot act without a minted token, and every minting is logged, so Vault always knows when one agent spawned another, a gap earlier research flagged[13].

03 **Data never becomes a command.** The master agent plans only from signed commands. Web pages, email and files go to a quarantined reader with no tools, which can extract values but cannot change the plan. Consequential arguments (a recipient, an amount, a file to share) must trace back to the human, not to something an agent read. In the CaMeL design this cut successful prompt injections from 300 of 949 to 0 on one model, while solving 77% of tasks against 84% undefended, at about 2.8× the tokens[1]. FIDES with policies cut them from 163 to 1[7].

04 **The gateway checks, not the agents.** Model-level defences keep falling: one 2026 study broke 12 of them, most at over 90% attack success[8]. So the checks run in deterministic code that no prompt can reach. That code verifies four things: that the chain leads back to the hub's key, that the calling agent holds the leaf token's key, that the arguments fit the token, and that consequential arguments came from the human.

05 **Everything is logged, including refusals.** Allowed, refused, escalated and orphan actions (no valid chain) are appended to a Merkle-tree log kept by the hub, the same structure as Certificate Transparency[9, 10]. Checkpoints are countersigned by a witness off the computer, such as the owner's phone[11], so a compromised machine cannot quietly rewrite history. An orphan action is the clearest alarm Vault has.

03 · WHEN YOU GET ASKED

When you get asked

Confirmation prompts work only if they are rare and honest, so the gateway asks only in specific cases (proposal). The text you confirm is generated by gateway code from a structured description of the call[17], not by an AI model. The hub shows or reads it, never the computer's own screen[6].

SITUATION	WHAT HAPPENS
Inside the command's scope, no untrusted data involved	Runs. Logged
Consequential (pay, delete, share outside, install) and inside scope	You confirm the plan once, then it runs step by step without further prompts
Outside the command's scope, or a consequential argument came from untrusted data	Suspended. The hub shows or reads the exact action with a warning such as "recipient came from an email". You re-verify; the hub mints a token for that one call
No valid chain to a verified command	Blocked, logged as an orphan, and you are alerted

There is no general standard for "what you see is what you sign" outside payments. WebAuthn dropped its transaction-text extension and Android deprecated its protected-confirmation screen[20, 21]. That is why the hub's own display and voice are the confirmation channel.

04 · ONE TRACE RECORD

One trace record

Every gateway decision produces one record, signed by the gateway, not by the agent (proposal, adapted from[2, 9, 18]). The log stores hashes; the content stays encrypted on the owner's device and can be revealed selectively.

PART	CONTENTS
Lineage	Record ID; root command ID; parent records (the plan steps and delegations that led here)
Human root	Owner ID; hash of the exact transcript; the hub's verification evidence for that utterance or typing window (channels, scores, times); hub signature
Authority	Token chain from root to leaf; each hop's agent identity (agent ID and signed binary hash); depth; effective permissions; the acting agent's proof of key possession
Action	Tool; hash of the arguments; where each argument came from (human, tool, or untrusted data)
Decision	Policy version; allowed, refused or escalated; if escalated, the confirmation and a hash of the exact text shown
Outcome	Output hash; start and end time on the hub's clock
Log position	Leaf index and the checkpoint current when it was written

What exists and what Vault adds

PIECE	WHAT EXISTS	WHAT VAULT ADDS
A signed human root	AP2 has a non-agentic surface sign the user's payment mandates, for payments only[6]; South et al. propose a human-signed delegation token, as research[5]	A root signed by the hub after continuous voice and hand verification, per command rather than per session
Narrowing tokens	Biscuit (shipping)[3], UCAN (release candidate)[4], attenuating agent tokens (IETF individual draft)[2]	The root key bound to a verified human
Injection resistance	CaMeL and FIDES planners with provenance labels[1, 7]	Enforcement in the gateway, outside any model
Agent protocols	MCP forbids passing tokens through, so command lineage does not travel hop to hop[15]; A2A can escalate a request up a chain of tasks to a human[16]	A chain token carried alongside MCP and A2A calls, checked by the gateway
Tamper-evident logs	Certificate Transparency[9], Sigstore Rekor v2[10], off-device witnesses[11]	Per-action records tied to a human root
Confirmation	Secure Payment Confirmation, for payments only[12]	The hub as the trusted display and voice for every consequential action

Open problems

- **No adaptive-attack test of the planner designs.** CaMeL and FIDES have not yet faced attackers who adapt to them[1, 7, 8].
- **Turning speech into scope.** The master agent's interpretation of a command is a trust boundary. The read-back reduces this risk but adds friction; how often owners actually listen to it is unmeasured[5].
- **Voice as the confirmation channel.** "What you hear is what you sign" has no standard and no security analysis.
- **Agents outside the computer.** Cloud agents only stay in the chain if the services they call check Vault tokens. Until then, their actions are visible only where they pass through the gateway.
- **Utility cost.** About 7 percentage points of task success and about 2.8× tokens for CaMeL-style planning[1].

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Hardware spec

The five devices Vault makes: what each one senses, the chips inside, how fast it samples, how it is powered, how it signs what it sends, how it pairs with the hub, and how to prototype it from parts on sale today.

Prepared by Pablo Peniche **Status** Draft for discussion · not for distribution **Sources** 56, numbered inline

Also the rig · the command chain · the P0 bench

IN ONE PARAGRAPH

Five devices, one hub, every signal signed.

Vault makes five devices: wired earbuds, two muscle-sensing wristbands, a keyboard, a camera module and a hub. Every device carries its own secure element and clock. It timestamps every sample, signs what it sends, and pairs only with its owner's hub through mutual attestation. Raw biometric signals go only to the hub, never to the computer. The earbuds, keyboard and camera are cabled to the hub. Only the wristbands use radio, one Bluetooth link per band. The parts below are candidates checked against datasheets, not final selections. Each device has a prototype path that starts with hardware on sale today.

Figures come from datasheets and vendor pages. Power and bandwidth numbers marked "estimate" are this spec's own arithmetic. "Proposal" marks design choices made for Vault.

01 · SHARED RULES

What every device does

RULE	WHAT EVERY VAULT DEVICE DOES
Identity	A secure element holds a device key, provisioned at the factory, with a certificate from a Vault certificate authority. Pairing is mutual attestation between device and hub, followed by an encrypted session, the way Apple's Touch ID keyboard pairs with a Mac[1]. A standard for authenticating USB devices exists, but no hub, host controller or operating system was found that supports it, so Vault runs its own[2]
Signing	Every data frame carries a message authentication code (MAC) under the session key, and the device signs a checkpoint every second (proposal). Signing each keystroke or audio frame individually may be too slow for a secure element; per-signature timing was not found
Time	Every sample is timestamped on the device. The hub disciplines all clocks to within 1 ms, because the voice check needs ±5 ms and the keystroke check ±50 ms (proposal). The hub's reference is a temperature-compensated oscillator accurate to ±0.1 ppm[3]
Firmware	Secure boot, signed updates, a rollback counter, and debug ports disabled after manufacturing (proposal)
Data	Raw audio, muscle signals and camera frames go only to the hub, encrypted. The computer receives keystrokes, transcribed text and check results, never raw streams
Links	Cables wherever possible, because Bluetooth relays survive encryption and add only about 8 ms[4]. The wristbands are the only radio devices

02 · EARBUDS

Earbuds

FUNCTION	SPEC	CANDIDATE PARTS
Skull-vibration sensor	8 or 16 kHz on the same audio bus and clock as the microphones, so the voice check is sample-locked by design	ST LSM6DSV16BX: bone channel over TDM, 30 µg/√Hz noise, plus a motion sensor[5]. Second source: Bosch BMA580, up to 6.4 kHz on its own clock[6]
Microphones	One inner and one outer per earbud, 48 kHz	Infineon IM69D129F: 69 dB(A) SNR, 450 µA[7]. TDK T5848 for the inner mic, with 133 dB SPL overload next to the speaker[8]

FUNCTION	SPEC	CANDIDATE PARTS
Speaker	Normal audio, plus inaudible ear-canal probes in speech pauses for identity[9]	Balanced-armature or dynamic driver
In-ear detection	Contact and motion from the sensor above, plus a capacitive pad (proposal)	Built into the motion sensor; capacitive pad on the shell
Body-code transmitter	Electro-quasistatic signal in the skin at 100–500 kHz. Lower carriers leak less: under 10 cm at 100 kHz against about 40 cm at 1 MHz[10]. Coherent modulation at 10–100 kb/s or more. Current of a few microamps[11]	Discrete DAC and electrode driver; Microchip shipped a discrete 125 kHz body-coupling kit in 2013[12]. Ixana's Wi-R chip advertises 100 kb/s–5 Mb/s, but its carrier is unpublished[13]
Security	Device key, per-frame MAC, checkpoint signatures	Microchip ATECC608B: P-256, 2×3 mm, under 150 nA asleep[14]
Link and power	USB-C to the hub: USB audio plus data. The cable also carries a shared ground that closes the body-code loop	Bus-powered with no battery: under about 150 mW against 2.5 W available (estimate)

The skull-versus-mic matching runs in the hub, on signed streams that left the earbud on one clock, so the earbud itself stays simple (proposal). The cable carries four 48 kHz microphone channels plus the skull channel, roughly 3 Mb/s of raw 16-bit audio (estimate), well within USB 2.0.

Prototype path.

01 Prove skull-versus-mic matching on OpenEarable 2.0, an open research earbud with a bone sensor and inner and outer mics[15].

02 Bench-test the body code: a signal generator and electrode at the ear, receivers at the wrist and on a keycap, measuring signal level and leakage at 100, 250 and 500 kHz.

03 Build a first wired earbud on a USB audio board with the LSM6DSV16BX and two mics.

Wristbands

FUNCTION	SPEC	CANDIDATE PARTS
Electrodes	16 bipolar channels from dry gold-plated electrodes (48 pins in Meta's design)	Meta's research band used about 10 pF in parallel with 10 TΩ input impedance, built from discrete parts in each electrode pod[16]
Front end	16 channels at 2 kHz, noise near 2.5 µVrms	Intan RHD2216: 16 differential channels, 2.4 µVrms, about 3.5 mW for 16 channels at 2 kS/s[17]. Fallback: two TI ADS1298, about 1.1 mW per channel[18]. Avoid the ADS1299 for all-day wear, at about 4.9 mW per channel[19]
Resolution	About 11 effective bits, since Meta's chain had about 65.5 dB of dynamic range[16]. Pack samples to 12 bits: 384 kb/s per band (estimate)	—
Motion	6-axis inertial sensor for context and the "wrist moved" cue	LSM6DSV16BX-class part[5]
Body-code receiver	A separate high-impedance electrode, an ADC and a digital correlator that hears the earbuds' code all the time (proposal)	Same front-end approach as the keycaps (section 4)
Skin contact	Electrode-impedance checks detect removal (proposal)	—
Radio and compute	Bluetooth LE 2M, one link per band, about 1.3–1.4 Mb/s at best[20]	Nordic nRF54L15: 2.45×2.25 mm, 4.8/3.4 mA transmit/receive at 0 dBm, TrustZone and a hardware key store for Ed25519 and P-256[21]
Security	Device key and per-packet MAC	The nRF54L15's key store, or an NXP SE050 (EAL6+, Ed25519) for a separate certified chip[22]
Battery	About 12–22 mW average with the RHD2216 and nRF54L15 streaming, so 18 hours needs about 75–140 mAh (estimate)	150–250 mAh pouch cell; a 150 mAh cell is about 20×26×3.8 mm[23]

Meta's research band ran about 4 hours per charge[16], and its consumer Neural Band about 18 hours[24]. Vault's target is a full working day, with a second pair swapping in during a fingerprint touch. The RHD2216's input impedance is about 13 MΩ at 1 kHz, far below the input stage of Meta's research band[17, 16]. Dry electrodes will probably need a buffer at each electrode, which is the main analog design risk.

Prototype path.

01 Develop the muscle-to-keystroke pairing on two MindRove armbands (8 channels at 500 Hz)[25] and an analog keyboard.

02 Build a tethered 16-channel band: an RHD2216 evaluation board with a dry-electrode strap.

03 Build the wireless band with the nRF54L15 and a 150–250 mAh cell.

04 · KEYBOARD

Keyboard

FUNCTION	SPEC	CANDIDATE PARTS
Switches	Analog Hall-effect switches, every key scanned at 8 kHz, actuation set in 0.1 mm steps; the Wooting 80HE ships this today[26]	TI TMAG5253 Hall sensor: 15 kHz bandwidth, 2 mA active[27]. Firmware base: libhmk, open source, 8 kHz[28]
Timestamps	Taken at actuation on the keyboard's own clock, before USB	Keyboard microcontroller
Electrode keycaps	Conductive keycap tops on spring contacts. When a key's Hall sensor shows travel, its electrode is routed to one of a few receive channels, so only pressed keys are listened to (proposal)	Per-key buffer: TI OPA320, 0.9 pA bias[29]. Low-capacitance mux: ADI ADG1208, 1 pF[30]. Correlator on an ST STM32G474 with five 4 Msps ADCs[31]. DiamondTouch used 100 kHz bursts and synchronous demodulation[32]
Fingerprint key	Encrypted link to the keyboard's secure element; matching for identity, confirmations and a duress finger	Fingerprint Cards FPC2534 AllKey Pro: AES-GCM host link, Ø11 mm, 508 dpi, false-accept rate 1 in 100,000, identification in 160–2,050 ms[33]. The development kit lists at $170, restock estimated January 2027[34]
Security	Device key; MAC on every key event; signed checkpoints	NXP SE051: EAL6+ (AVA_VAN.5), about $2.88 at 1,000 units[35, 36]
Link	USB-C to the hub only, never directly to the computer. The hub presents an ordinary keyboard to the computer once the checks pass	Keyboard microcontroller with USB device mode

The fingerprint key needs its own trust design. Windows Enhanced Sign-in Security accepts only match-on-sensor parts certified with Microsoft at manufacture, which a small builder can't realistically get[37]. In 2023, researchers bypassed shipping laptop fingerprint sensors where the secure channel was misconfigured or missing[38]. The FPC2534 has an encrypted link but no attestation, and its lockout resets with its firmware[33]. So the keyboard pairs the sensor at the factory, keeps its own attempt counter, and attests to the hub, Apple-style (proposal)[1].

Prototype path.

01 Wooting 80HE plus its analog SDK, for key timing[26].

02 One hand-made copper-foil electrode keycap on a bench receiver, to measure body-code detection per press.

03 A custom keyboard PCB on libhmk with electrode keycaps, an SE051 and an FPC2534 fingerprint key.

05 · CAMERA MODULE

Camera module

FUNCTION	SPEC	CANDIDATE PARTS
Image sensor	Monochrome global shutter, about 1 MP, 60–120 fps	OmniVision OV9282: 1280×800 at 120 fps, 2-lane MIPI, in volume production[39]. ams OSRAM Mira220: 1600×1400 at 90 fps, quantum efficiency 56% at 850 nm and 36% at 940 nm[40]
Illumination	Its own 850 or 940 nm light, so it works in the dark, pulsed in a random pattern per frame as a liveness challenge (proposal)	VCSEL emitter classified eye-safe (Class 1) under IEC 60825-1[41]
Depth for liveness	A coarse 3D check that rejects flat images	ST VL53L8CX: 8×8 zones at 60 Hz, 940 nm, Class 1[42]. For full depth, Infineon IRS2976C time-of-flight, which passed Google's strongest face-unlock class[43]
Processor	Detects and crops the face, counts people in view, MACs every frame under the hub session key and signs checkpoints, with a DICE-derived device identity (proposal)	ST STM32N6: MIPI camera input, image signal processor, 600 GOPS NPU[44]

FUNCTION	SPEC	CANDIDATE PARTS
Bandwidth	Full frames at 120 fps are about 983 Mb/s, beyond USB 2.0 (estimate), so the module sends the cropped face region or uses USB 3	—
Link	USB-C to the hub; frames never reach the computer	—

A custom module is needed because Windows Enhanced Sign-in Security doesn't support external cameras[37]. MIPI has published camera security specifications, but they are members-only, and no sensor advertising them was found[45]. The signing therefore happens in the module's processor rather than in the sensor itself.

Prototype path.

01 A USB infrared camera on a single-board computer standing in for the module.

02 An STM32N6 development kit with an OV9282 sensor, to prove frame signing and on-module cropping.

06 · HUB

Hub

FUNCTION	SPEC	CANDIDATE PARTS
Processor	Runs voice matching, keystroke decoding and face matching in an isolated environment	NXP i.MX 93, NPU of about 0.5 TOPS[46]; or i.MX 95, about 2 TOPS and a 10–15-year supply program[47]. Both include NXP's EdgeLock secure enclave with tamper support
Secure element	Root keys, device certificates, signatures on the trace log	NXP SE051, EAL6+ (AVA_VAN.5)[35]
Radios	One Bluetooth receiver per wristband, so two bands never share a link	Nordic nRF54L15[21]
Ports	USB-C downstream to the earbuds, keyboard and camera; USB-C upstream to the computer, which sees an ordinary keyboard and audio device only once the checks pass	—

FUNCTION	SPEC	CANDIDATE PARTS
Clock	Disciplines every device to within 1 ms	SiTime SiT5356 temperature-compensated oscillator, ±0.1 ppm[3]; RV-3032-C7 real-time clock, ±1.5 ppm[48]
Tamper	Opening the case erases the keys	Enclosure mesh plus ADI DS3645 battery-backed key memory with mesh inputs[49]
Trusted display and speaker	Shows or reads the exact action to confirm, driven only by the hub's secure side; see the command chain	Small display and speaker (parts not yet selected)
Trace log	Append-only Merkle log in encrypted flash, with checkpoints witnessed by the owner's phone	—
Power	USB-C Power Delivery from the wall	—

Prototype path.

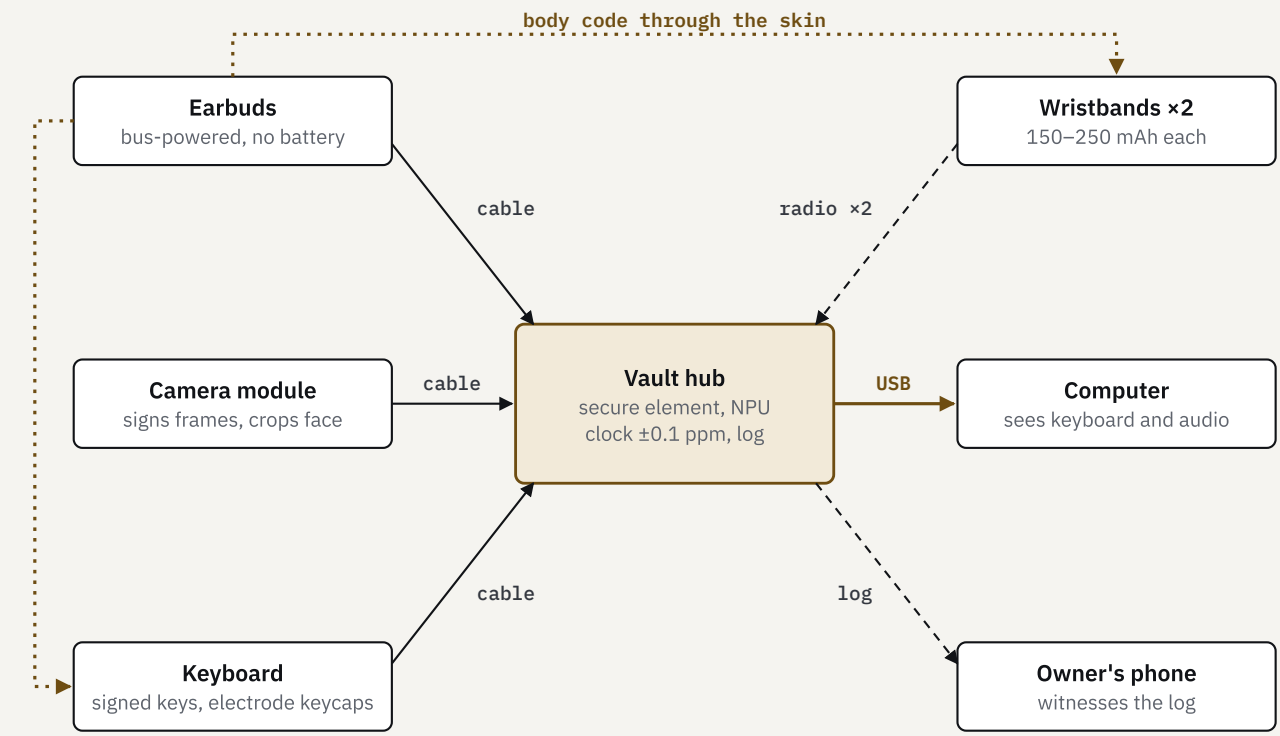
01 Raspberry Pi Compute Module 5, from $67.50[50], with the AI HAT+ 2 (40 TOPS INT4, $200)[51] and an SE051 evaluation board. The Pi has no trusted execution environment, so this stage proves the checks, not the security.

02 An i.MX 93 evaluation board with its secure enclave, the SE051 and the nRF54L15 receivers.

07 · LINKS, CLOCKS, KEYS

Links, clocks and keys

Three cables and two radio links reach the hub. The body code travels separately, through the skin, from the earbuds to the wristbands and the keycaps.



How the devices connect (proposal). Three cables and two radio links reach the hub, and only the wristbands use radio[4]. The computer sees an ordinary keyboard and audio device, and only once the checks pass. The body code (dotted) never goes through the hub: it runs through the skin, from the earbuds to the wristbands and the keycaps, which report what they heard[10, 11].

DEVICE	LINK TO HUB	DATA	CLOCK SYNC (PROPOSAL)	KEYS
Earbuds	USB-C cable	4 mics at 48 kHz plus skull channel, about 3 Mb/s raw (estimate)	USB frame timing	ATECC608B device key; session MAC
Wristbands	Bluetooth LE 2M, one receiver per band	16 channels × 2 kHz × 12 bits = 384 kb/s per band (estimate)	Hub time beacon at each connection event	nRF54L15 key store
Keyboard	USB-C cable	Key events with depth and keycap body-code results; encrypted fingerprint data	USB frame timing	SE051 device key; MAC per key event
Camera module	USB-C cable (USB 3, or cropped frames)	Cropped face region at 60–120 fps	USB frame timing	DICE identity in the STM32N6
Hub to computer	USB-C upstream	Keyboard and audio only after checks pass	—	—

08 · PROTOTYPE PATH

Prototype path

PHASE	BUILD	WHAT IT PROVES	PASS MARK (PROPOSAL)
P0, bench	OpenEarable 2.0, two MindRove bands, a Wooting 80HE, a copper-foil electrode keycap, a signal-generator body code, and a Raspberry Pi hub; the bench plan prices and wires the build	The three core checks work at all	Dictation from the wearer accepted at least 97% of the time with loudspeaker playback rejected, as in VAuth[52]; at least 95% of keys paired in 20-key windows[53]; body code detected on at least 99% of presses at a few microamps
P1, device boards	Wired earbud board; tethered RHD2216 band; keyboard PCB with electrode keycaps, SE051 and FPC2534; STM32N6 camera; i.MX 93 hub	Each device signs its data and pairs with the hub	Every injected frame, key or audio packet rejected; clocks within 1 ms
P2, alpha	Wireless bands with nRF54L15 and 150–250 mAh cells; enclosures; tamper; trace log; trusted display	A full working day of use	A full day on one charge; false lockouts rare enough to keep working (to be measured)

09 · OPEN QUESTIONS

Open questions

- **Skin current.** Body-channel currents of a few microamps sit about 2,000 times below the contact current where pain begins[11, 54]. Continuous all-day injection still needs a medical-safety review, including for people with pacemakers.
- **Dry electrodes on the RHD2216.** Its input impedance is far below Meta's front end, so per-electrode buffers may be needed[17, 16].
- **Signing speed.** How long a secure element takes per signature decides whether every event can be signed or only MAC'd with periodic signed checkpoints.
- **Body-channel chips.** Ixana publishes no carrier frequency or price, and reported 100,000-unit minimum orders in 2023[13, 55]. A discrete transmitter and receiver is the safer plan.
- **Shared ground.** Wired earbuds and the keyboard share the hub's ground, which changes body-code signal strength and leakage compared with the floating setups studied. An Oxford prototype told the body channel from fakes 99.3% of the time, with its lab instruments decoupled from ground to simulate battery-powered devices[56]. Measure on the bench.
- **Fingerprint supply.** The FPC2534 development kit is out of stock until about January 2027[34].

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The P0 bench

The first build phase, priced and wired: what to buy, how the bench connects and syncs, the software behind each check, the phantom-based body-code tests, and the protocol behind every pass mark.

Prepared by Pablo Peniche **Status** Draft for discussion · not for distribution **Sources** 28, numbered inline

Also the rig · the hardware spec · the command chain

IN ONE PARAGRAPH

One bench, about $5,000, and the first numbers for the three checks.

The P0 bench is the first stop on the prototype path: one table of off-the-shelf parts, no Vault-made hardware, no secure elements, nothing a deployment would trust. Research earbuds stand in for the earbuds, two muscle-sensing armbands for the wristbands, an analog keyboard for the keyboard, a signal generator and a microcontroller board for the body code, and a Raspberry Pi for the hub. The bench exists because none of the three core checks has been measured: skull-versus-microphone voice matching against replay, per-key pairing of muscle bursts with key presses, and a body code carried from the ear to the wrists and the keycaps. An agar-saline phantom stands in for the owner during the body-code sweeps, so those numbers repeat day to day, and short live runs on the owner keep them honest. Everything is on sale today, the two research pieces arrive made-to-order in two to three weeks, and the whole list costs about $5,000 before tax and shipping.

Prices are vendor prices checked in October 2026, before tax and shipping. The earbuds are quoted in euros and converted at about $1.12. "Proposal" marks design choices made for the bench, and the pass marks are proposals carried over from the hardware spec, not established benchmarks.

01 · THE BENCH

What P0 measures, what it defers

P0 answers one question: do the three core checks work at all, on parts you can buy this month? The hardware spec puts it first in the build plan for exactly that reason. Everything after it, the device boards of P1 and the alpha rig of P2, assumes these numbers exist.

CHECK	QUESTION	P0 PASS MARK (PROPOSAL)
Same voice	Did the wearer's own throat produce this speech?	Dictation from the wearer accepted at least 97% of the time, loudspeaker replay rejected
Same hands	Did the owner's own fingers press these keys?	At least 95% of keys paired in 20-key windows
Body code	Is the earbud's signal on the body that pressed the key?	Detected on at least 99% of presses at a few microamps

What P0 defers is everything that makes the rig a product. No device signs anything, because the Raspberry Pi hub has no trusted execution environment and no secure elements are on the bench. The camera and the face check are P1 work, and the fingerprint key waits anyway: the FPC2534 development kit is listed at $170 with restock estimated January 2027[1]. The bench also measures something the spec asks for but does not have: how much a shared ground changes the body channel, since the published body-channel prototypes all floated their loops, decoupling lab instruments from ground to simulate battery-powered devices[2]. One owner and two colleagues are enough wearers for P0; the pass marks are per-owner numbers, and VAuth's 18-user study remains the reference scale[3].

02 · SHOPPING LIST

Every part, priced

Three groups: the stand-ins for Vault's devices, the bench instruments that drive and measure the body code, and the consumables. Prices checked October 2026.

Stand-ins for the rig.

PART	STANDS IN FOR	WHY THIS ONE	PRICE
OpenEarable 2.0, developer bundle (a left and right pair)	The earbuds	The open research earbud: inner and outer microphones plus a 3-axis bone-conduction sensor, recorded on one device clock to microSD, mics at up to 48 kHz and the bone channel at up to 6.4 kHz[4, 5]. BLE streaming tops out at 800 Hz, so the bench records to card	€2,348, about $2,630; made to order, ships in 2–3 weeks[6]
MindRove armband ×2	The wristbands	8 channels of raw sEMG at 500 Hz per band (0–250 Hz bandwidth, 24-bit), 8+2 stainless electrodes, Wi-Fi to the hub, Python SDK on Linux; 4–6 hours per charge[7]	$679 each on current sale (lists at $729), $1,358
Wooting 80HE	The keyboard	Analog Hall-effect switches, every key scanned at 8 kHz, 0.1–4.0 mm travel reported per key, with an open analog SDK[8]	$199.99
Raspberry Pi Development Kit for Compute Module 5 (4 GB, wireless)	The hub	The spec's prototype hub in one box: the module, IO board, power supply, cooler and antennas[9]	$195

Bench instruments for the body code.

PART	ROLE	WHY THIS ONE	PRICE
Digilent Analog Discovery 3	The body-code transmitter, and the bench meter	A 2-channel 14-bit waveform generator and 2-channel 125 MS/s oscilloscope on USB: one AWG drives the ear electrode, the other channels measure levels and leakage, and WaveForms scripts it all from Python — including a software lock-in amplifier for the small carrier signals[10]	$379
NUCLEO-G474RE	The receiver for the wrist electrode and the electrode keycap	The spec's keycap correlator chip on a $20 board: five 12-bit ADCs at 4 Msps and a CORDIC unit for the matched filter[11]	$20
Two OPA320 buffers on SOT-23 adapters, plus passives	The receiver front end	0.9 pA input bias keeps the electrodes high-impedance, the same part the spec puts behind each keycap[12]	About $25 (estimate)

Electrodes, the phantom, and small parts.

PART	ROLE	NOTES	PRICE
Kendall Ag/AgCl foam electrodes, 50-pack ×2	Skin electrodes: the ear transmitter, the wrist receiver, the sync pulse	Pre-gelled, snap connectors, the usual research-grade consumable[13]	$24.99 each, $49.98
Copper tape, wire, BNC leads, breadboard	The electrode keycap and the harness	A stock keycap wrapped in copper foil over a spring contact, wired to the receiver: the hand-made keycap the spec calls for	About $60 (estimate)
Agar, salt, PVC pipe, two containers, gloves	The body phantom (section 5)	Agar-saline mixed to the muscle range; agar phantoms hold conductivity between about 0.02 and 1.23 S/m and keep their shape for a year[15, 16]	About $60 (estimate)
Conductivity meter, piezo sounder, finger cot, acrylic rod	Phantom checks, the sync chirp, the two press tools	The meter verifies each phantom mix before a sweep; the agar-filled cot and the rod are the paired press test of section 5	About $50 (estimate)
Any powered 3.5 mm speaker	The loudspeaker attack	Replays the owner's recorded voice at the wearer	About $30 (estimate)
microSD cards, USB-C audio dongle, spares	Storage and replay plumbing	OpenEarable records to microSD and recommends the Micron 64 GB card; the dongle gives the hub an audio jack for replays[5]	About $60 (estimate)

The itemized total is about $5,100, of which the earbuds and the two bands are about $4,000. Not on the list on purpose: the SE051 evaluation board and all signing (P1), the camera module and the face check (P1), and the AI HAT+ 2 at $200[18], which joins when the camera and on-device transcription arrive. The Raspberry Pi alone runs all three P0 checks.

03 · WIRING AND SYNC

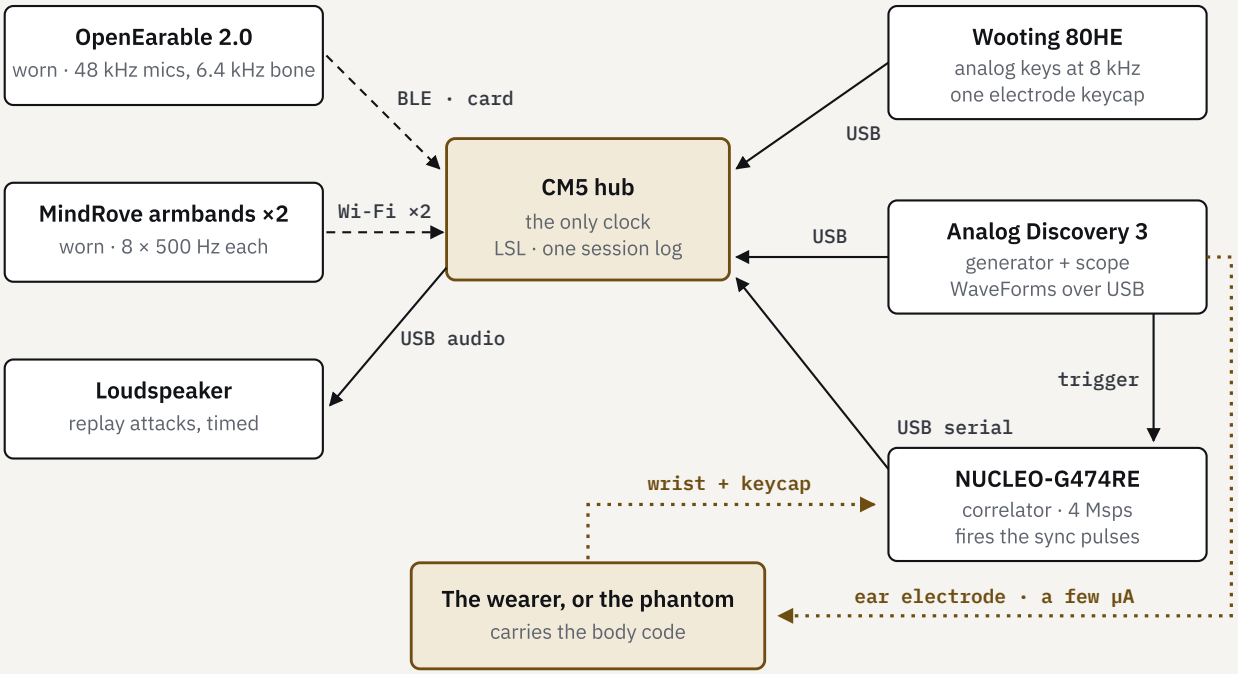
One clock, one shared ground

Everything lands on the hub, which is the only clock on the bench. The body code bypasses the hub entirely: it runs from the generator's electrode, through the wearer or the phantom, to the receiver's electrodes, exactly as it will

between the earbuds, the wristbands and the keycaps.

DEVICE	LINK TO THE CM5 HUB	WHAT FLOWS
OpenEarable 2.0	BLE while worn; microSD card pulled after each session	Sensor control and 800 Hz streams over BLE; the full-rate 48 kHz mics and 6.4 kHz bone channel arrive as files[5]
MindRove armbands ×2	Wi-Fi, one SDK connection per band	8 × 500 Hz sEMG plus IMU per band[19]
Wooting 80HE	USB	Analog key positions at 8 kHz, read through the analog SDK[20]
NUCLEO-G474RE	USB serial	Per-press correlation results, and the sync pulses it fires
Analog Discovery 3	USB, from the hub or from a battery laptop	The AWG out to the ear electrode; the scope back from the receivers and the leakage probe[10]
Loudspeaker	USB-C audio dongle	Replay attacks, timed from the hub so they land in the log

The body-code loop closes through ground, and the bench measures both configurations the spec asks about. With the AD3 on the hub's USB, the generator, the receiver and the keyboard share the hub's ground, which is what the product will do; with the AD3 on a battery laptop the loop floats, which is how the published prototypes ran, their instruments decoupled from ground to stand in for battery-powered devices[2]. DiamondTouch is the clear exception: its receivers shared a ground with the transmitter[14]. Section 5 measures the difference.



How the P0 bench is wired (proposal). Every stream lands on the CM5 hub, the only clock: USB for the keyboard, the generator and the receiver, Wi-Fi for the bands, and BLE plus a microSD card pull for the earbuds[5, 19]. The body code (dotted) never touches the hub: the generator drives an ear electrode at a few microamps, and the receiver listens at the wrist and the keycap[26]. A trigger line from the generator clocks the receiver so demodulation stays synchronous[14], and the receiver fires the sync pulses.

Sync is a measurement of its own. The earbuds and the bands cannot share the hub's clock natively, so the Nucleo fires a physical sync pulse every session start and every 60 seconds: it drives a piezo chirp pressed to the earpiece, which lands in both the microphone and the bone channel, and it injects a skin pulse near each band, which lands on all eight EMG channels. Post-hoc cross-correlation gives every stream its offset and drift against the hub, and the residuals are a reported result. The voice check needs ±5 ms and the keystroke pairing ±50 ms, and the bench targets residuals under 1 ms after correction. All streams ingest through the Lab Streaming Layer, so the analysis reads one timeline[21]. The Wooting needs no such help: its keys scan at 8 kHz, so a key-down lands on the hub within 0.125 ms of actuation.

04 · SOFTWARE PER CHECK

What runs where

One Python code base on the CM5, one process per source, everything written to disk per session: the raw streams, the parameters, and the firmware and script hashes. The pass-mark numbers in section 6 are all computed by the same offline code that the live path uses.

Same voice. The pipeline takes the microSD files: the inner microphone at 48 kHz and the bone channel at 6.4 kHz, resampled to a common rate. Voice activity detection finds the voiced 20–40 ms frames on the microphone; for each frame, the normalized cross-correlation between the two channels must peak within ±5 ms. That is VAuth's structure, feature matching between body-conducted and air-conducted speech, updated from their 2017 hardware[3]. An utterance is accepted when at least 80% of its voiced frames match (proposal), with the correlation threshold set per wearer on ten calibration utterances. Accepted audio is transcribed offline with Whisper to confirm the dictation itself survives the gate[22].

Same hands. The MindRove SDK streams both bands[19]; the Wooting analog SDK feeds key travel, with the tachywooting Python bindings logging per-key 0.0–1.0 positions with microsecond timing[20, 23]. Per band: bandpass 20–250 Hz, envelope, then burst detection with a Teager–Kaiser energy threshold. Pairing assigns each key-down exactly one burst that starts 80–200 ms before it, from the correct hand, with a plausible finger from the channel topography; the timing window comes from the forearm-muscle measurements Gazzari et al. report from Dennerlein's key-switch studies, where bursts begin about 150 ms before a key bottoms out[24]. The score is per 20-key window, because at typing speed the bursts overlap and even owner-trained decoders misread 7–11% of single characters, as emg2qwerty showed across 108 users[25]. A 200-key calibration run per owner fixes the channel-to-finger templates and the thresholds.

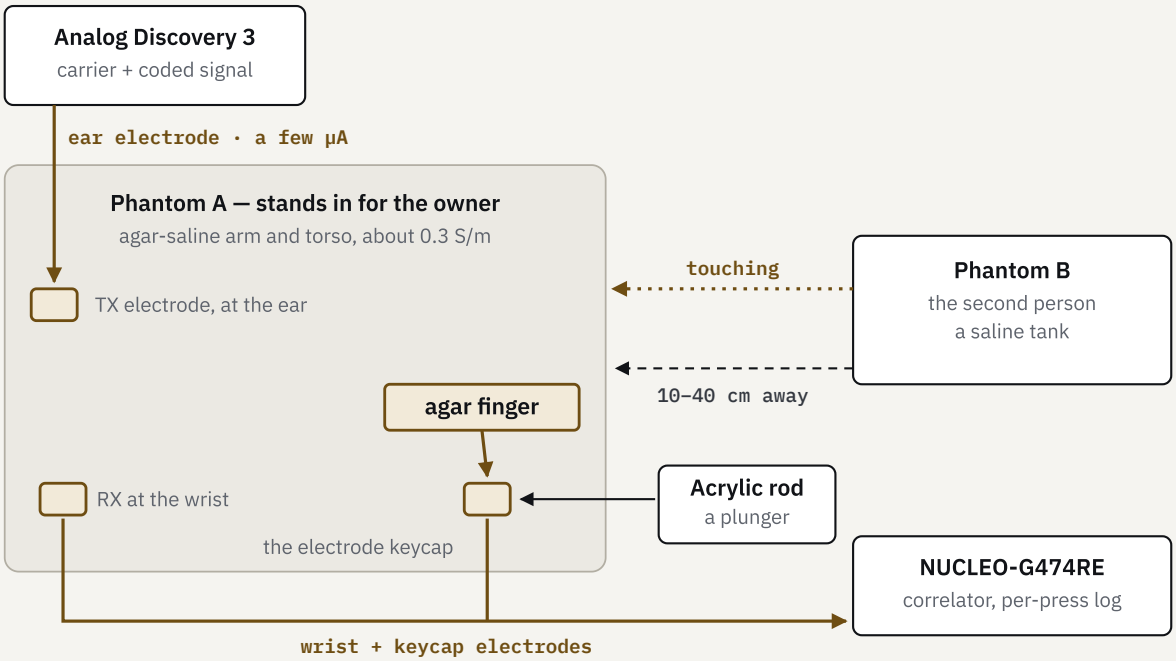
Body code. A WaveForms script drives the AWG: a carrier at 100, 250 or 500 kHz, on-off keyed by a 127-chip m-sequence at 10–100 kb/s, repeated throughout the press. Series resistance sets the skin current and the scope channel verifies it across a shunt, a few microamps, about 2,000 times below the contact current where pain begins[26]. On the receive side, each electrode feeds its OPA320 buffer into a 4 Msps ADC channel on the Nucleo, whose firmware mixes at the carrier and runs the matched filter against the known sequence, using the CORDIC unit; the AWG's trigger output clocks the Nucleo, so demodulation is synchronous, the DiamondTouch approach[14]. The Wooting's travel signal gates the keycap electrode: it is listened to only while a press is in flight. The Nucleo reports a correlation peak and a level per press, plus the sync pulses.

05 · PHANTOM TESTS

The body code without a body

The phantom exists for repeatability. A person's impedance, posture and sweat change hour to hour; an agar-saline phantom holds a set conductivity and geometry for a year[15, 16]. Intrabody-communication work already uses liquid phantoms of 0–1 S/m to cover the tissue range[17], so the bench builds one rather than inventing a new method.

Phantom A stands in for the owner: a torso block and a 60 cm arm tube of agar-saline mixed to about 0.3 S/m, inside the muscle range at these frequencies (proposal), verified with a conductivity meter before each sweep. The transmitter electrode sits at the tragus position, the receiver electrodes at the wrist position and under the keycap, at the same spacings a seated owner would have. Phantom B, a smaller saline tank, plays the second person. An agar-filled finger cot on a stick, coupled to the arm, presses the keycap; an acrylic rod presses it without coupling, the mechanical-plunger attack the rig page describes.



The phantom stand-in (proposal). Phantom A puts the ear-to-wrist channel at repeatable geometry and conductivity, mixed to the muscle range and checked before each sweep[15, 17]. The agar finger carries the code onto the keycap; the acrylic rod does not, and that difference is the per-press pass mark. Phantom B quantifies how much crosses to a touching stranger and how fast it fades with distance[28].

SWEEP	WHAT VARIES	WHAT IS RECORDED
Level	Carrier 100, 250, 500 kHz; skin current 1, 3, 10 µA; hub-shared ground vs floating	Signal at the wrist and keycap electrodes, as dB above the receiver's noise floor, with the current actually measured
Leakage	Probe electrode at 5, 10, 20, 40 cm off the body; phantom B touching vs not touching	Level vs distance, and whether the code crosses to a stranger through touch
Press	1,000 presses each: agar finger, acrylic rod	Detection rate with the code present vs absent, and the per-press margin
Rate	m-sequence at 10, 50, 100 kb/s	Correlation margin vs rate, and code errors

Leakage at 100 kHz should stay under about 10 cm, against about 40 cm at 1 MHz, which is why the carriers start low[27]. Touch is expected to carry the code to phantom B, a relay risk that body-channel defenses still struggle to bound[28]; the bench quantifies how much crosses, in dB, rather than assuming it.

Then the same levels, currents and press counts run live on the owner, one short session, same few microamps, same electrodes. The phantom buys repeatability and the owner buys validity, and the phantom-to-skin delta is itself one of the bench's reported numbers.

06 · PASS MARKS

How each one is measured

Every pass mark below is a rate with its raw counts and a 95% binomial interval, computed by the offline path, per wearer, from logged sessions. With 200 utterances the interval on 97% is about ±2.4 points, so the bench records margins, not just pass or fail (proposal).

CHECK	PASS MARK (PROPOSAL)	PROTOCOL	METRIC
Same voice	Dictation from the wearer accepted at least 97% of the time; loudspeaker replay rejected	3 wearers × 200 utterances of ordinary dictation; each wearer's 50 recorded utterances replayed from 30 cm at 75 dB and 90 dB; 50 utterances spoken live by each non-wearer while the wearer wears the earbuds	Acceptance: share of the wearer's utterances with ≥80% of voiced frames matching within ±5 ms. Rejection: share of replays and impostor utterances below that bar, VAuth's 97% and 0.1% as the reference[3]
Same hands	At least 95% of keys paired in 20-key windows	3 typists × 5,000 keys of ordinary prose at their own speed, after a 200-key calibration each; roughly 750 windows in total	Per non-overlapping 20-key window: the share of key-downs with exactly one burst 80–200 ms earlier, the right hand and a plausible finger; also the orphan-burst rate. Report the median and 10th-percentile windows per typist[24, 25]
Body code	Detected on at least 99% of presses at a few microamps	1,000 presses on the phantom with the agar finger; 1,000 live presses by the owner; 1,000 acrylic-rod presses as the negative control; all at ≤10 µA measured skin current, targeting 1–5 µA	Detection: correlation peak above a threshold fixed on 100 calibration presses. False positives: rod presses detected, target ≤1%. The level and leakage tables of section 5 are required artifacts alongside it[26]
Sync	Supporting number, not a pass mark	Sync pulses at session start and every 60 s, every session	Per-stream offset and drift after correction, and the worst-case residual, target under 1 ms

A check that misses its mark is a result, not a failure of the bench: it feeds P1 directly. If the MindRove bands miss bursts, that is evidence for the per-electrode buffers the spec already worries about; if the shared ground costs the channel tens of dB, the earbud's transmitter design changes before a PCB exists.

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Vault

A computer that only its owner, and the successors they name, can use. Not a login gate: an always-on loop of several channels at once that keeps proving a living human is at the keyboard, which human it is, and that every input comes from them. For everyday laptops and desktops, and for the terminals that guard frontier AI models.

Prepared by Pablo Peniche **Status** Draft for discussion · not for distribution **Sources** 279, numbered inline

Supersedes v1, the gate-based design (PDF)

BOTTOM LINE

Prove the owner behind every input.

Vault should be built as a closed verification loop whose output is not a session but authority to act, re-earned every few seconds and attributed to each input, and held only while one invariant is true: everything reaching the protected context is explained by a single attested, living body, that body is the enrolled owner or a successor the owner named, and no channel contradicts it. Actions by the owner's AI agents travel on a separate, owner-signed track. Nothing found on the market does this. Consumer platforms verify a paired token's proximity, its custody since the last unlock, or identity only at risky moments, and frontier labs verify tokens and second approvers[1, 2, 3, 4, 5]. The evidence supports a loop with five parts. On-body carriers prove liveness and keep custody. Attested device-side sensors supply identity. Ambient sensors report presence. Cross-checks run by the trusted channel bind these sources together. A trust engine lets every live channel veto but no channel vouch alone. The loop is enforced outside the general-purpose operating system: by a secure-element hub on consumer machines, and by a secretless terminal feeding a confidential-computing keyserver at AI labs. The founder's two instincts survive in modified form. "All channels must be on" fails as a check-by-check AND, because five channels that each falsely reject 1% of the time, checked every 10 seconds, would falsely fail about 141 times in an 8-hour day (this report's arithmetic, Section 7). It works as "every proposition is covered at every moment and any channel can veto". "A blood sample" cannot identify anyone continuously, but a single-use microneedle sensor is the strongest custody seal available[6, 7]. The loop cannot see coercion, so duress and succession remain ceremonies. And proof that an input came from a human hand, not an agent or an injected device, is the least mature link on both tiers.

Labelling conventions used throughout: "vendor claim" marks a figure published by a vendor and not independently evaluated; "secondary" marks press, aggregator or search-summary coverage that was not checked against the primary source; "proposal" and "inference" mark design reasoning produced by this research rather than established fact.

Verification becomes a loop that leases authority per input

A gate asks one question at the door, "may this person begin?", and then trusts whatever happens inside. Keeping the gate open with periodic presence attestations improves it but does not change the question being answered. Each attestation says that an enrolled face is still in front of the camera. It says nothing about whether the keystrokes now arriving came from that person's hands, from the person's AI agent, from malware riding the session, or from a cable that pretends to be a keyboard. It says nothing either about whether the person in the chair is acting freely. At least four of the threats analyzed in Section 2 (AI agents, injected input, piggybacked malware and coercion) can operate while a legitimate owner sits verified at the machine. A presence-kept gate passes all four by construction.

The thesis of this report is that Vault must instead be a **control loop**. In a loop, verification is a continuously evaluated invariant, and authority is a quantity the system modulates moment by moment, not a state it grants once. The loop continuously tests four propositions. The first is that a **living human body is present and attached** to the machine's sensors. The second is that this body is **the enrolled person**. The third is that **the inputs come from that body** and not from an agent, malware or injected device. The fourth, tested separately because nothing can measure it directly, is that **the person is acting freely**. Channels on the body, on the device and in the room produce evidence. Cross-checks bind that evidence to one body. A trust engine turns it into a running estimate that the invariant holds. An enforcement point outside the operating system converts the estimate into keys, input routing and capability, epoch by epoch. When custody breaks, the loop does not reopen a door; it schedules its own re-anchoring check. When channels contradict each other, authority ends at once. The owner's agents are separate principals with their own identity and owner-signed scope.

Eight findings support the thesis:

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01 **No shipping platform verifies identity continuously in hardware; each verifies something weaker.** Apple Watch unlock verifies custody of a token since the last real unlock within 2–3 m. Android Extend Unlock caps unlocked time at 4 hours, and its on-body mode cannot tell who carries the phone. Apple Stolen Device Protection and Android Identity Check re-verify identity only at risky moments. Nymi authenticates once per shift and carries trust on on-body detection (vendor claim)[1, 2, 3, 8, 9]. Lab frameworks commit to MFA, hardware keys and two-party approval, not to a continuously verified human[4, 5, 10]. Enterprise agents that run periodic face or behavioural checks exist, but as endpoint software without published liveness or injection defences (vendor claims)[11, 12]. Vault would be the first consumer product to attempt continuous identity, and the first found anywhere to root it in hardware (inference from this search).

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02 **The channels split the work unevenly: liveness is cheap, identity is scarce, custody is negative evidence, and input provenance barely exists.** Pulse, radar and interstitial-fluid sensors readily show a living body. Identity is reliable only from a few hard biometrics on attested paths. Wrist PPG identification collapsed to a Matthews correlation coefficient (MCC) of 0.07–0.08 when trained on one half of a semester and tested on the other[13], and the Nymi Band's ECG check was spoofed 81% of the time[14]. Custody always means "no break was detected", never positive proof. No shipping keyboard signs ordinary keystrokes[15].

03 **Agreement between channels binds them to one body but never identifies that body.** Real-time face swaps carry the pulse of the person driving them, with r = 0.89 for DeepFaceLive[16]. Tools exist to insert a chosen pulse into video[17]. Wrist-to-keyboard correlation (ZEBRA) was evaded about 40% of the time, because the attacker's own input decided when comparisons happened[18, 19]. Identity must enter through a hard biometric on an attested path and then be carried by custody, and the trusted channel must drive every comparison.

04 **"All channels must be on" fails as a per-check AND and works as universal veto plus role coverage.** Five channels at a 1% false-reject rate fail 4.9% of checks together. Daugman shows that combining a weak test with a strong one by AND at their crossover points can be 5.5 times worse than the strong test alone[20]. Requiring three consecutive failures cuts the 141 daily false lockouts to about 0.34 at 30 seconds of added latency for *missing* evidence. Treating *contradictions* as instant vetoes keeps impostor detection immediate (inference from the arithmetic in Section 7).

05 **A blood sample cannot identify anyone continuously, but a microneedle sensor is the best custody seal available.** Rapid DNA takes about 90 minutes, and DNA is shed everywhere[6, 21]. Metabolic profiles uniquely re-identify only about 40% of people after seven years and need laboratory instruments[22]. A single-use microneedle sensor that "cannot be re-applied" once it falls off[7] and streams live interstitial chemistry for 5–15 days is harder to transfer than any clasp. That holds only if it is applied during a strong identity ceremony, and no glucose-sensor maker documents a security-grade removal event today[23].

06 **The AI-agent question is answered by attribution, not detection.** Operating-system injection flags only describe which software path an event took. A kernel virtual-HID driver or a malicious USB cable looks like a keyboard[24, 25], and desktop agents use the same permission as malware[26, 27]. The loop should attribute each input to the present owner, to the owner's agent (a distinct account holding an owner-signed mandate), or to nobody, in which case it is blocked. It should then confirm consequential agent actions through a channel the agent cannot press[28, 29, 30].

07 **Enforcement has to sit outside the general-purpose OS on both tiers.** Security tools were disabled in about 48% of 2024 intrusions[31], and a local administrator could inject Windows Hello face templates[32]. For labs, the secretless terminal and confidential-computing keyserver remain the enforcement point, now fed signed loop state instead of presence heartbeats[33, 34]. The consumer equivalent this report proposes is a small hub with a secure element that terminates the wearable links, sits inline between keyboard and computer, and holds the vault keys. The add-on sensor kit costs roughly $640–$1,400, often on top of hardware buyers already own (inference from listed prices).

08 **The loop cannot see coercion, so duress and succession stay ceremonies.** Stress inference fails statistically, and for EU employees it is presumptively prohibited[35, 36, 37]. Explicit covert signals, silent degradation, quorum and delay remain the answer[38, 39, 40]. The loop helps in two narrow ways: it offers more covert signalling surfaces, and its owner-liveness signal can hold the heir path closed. The stakes are rising. Verified crypto "wrench attacks" hit a record 72 in 2025[41], and a computer bound to one body makes that body the attack surface.

The two tiers share one architecture and differ in enforcement, law and buyers. A consumer Vault runs on commodity laptops and desktops, faces US consumer health-data law (Washington's My Health My Data Act counts keystroke rhythms and vital signs)[42], and sells to high-risk individuals through custody, executive-protection and family-office channels. An AI-lab Vault runs on dedicated terminals, faces the EU AI Act's workplace emotion-recognition ban and employment biometric law, and sells to an estimated 5–12 frontier developers plus their infrastructure partners (an inference built on Epoch AI's count of 12 developers with models above 10^25 FLOP)[43]. Sections 8, 11 and 12 treat both tiers side by side.

The dangerous moments come after the check

A gate verifies at an event and then trusts a session. Even a gate that renews itself every few seconds with a signed "enrolled face present" attestation answers a narrow question: whether the right person is nearby. The threat record shows that the costly failures happen inside the session, through paths that this question never covers.

Handover after login is the oldest gap and the shipping precedents leave it open

Every consumer continuity feature leaves a window in which someone else can take over an unlocked machine. Android's on-body detection may take "up to one minute" to lock after a phone is set down, and "5-10 minutes" after the owner gets into a vehicle. Its trusted-device mode lets anyone near a paired Bluetooth device keep the phone unlocked, and Google itself warns that a Bluetooth connection can be imitated[2]. Reviewers measured 15–20 seconds to relock, and a 2019 bug left phones unlocked for minutes on a still table[44, 45]. Windows Dynamic Lock locks "within about a minute" of the paired phone leaving range, and never unlocks[46]. Windows presence sensing spaces locks at least 15 seconds apart to prevent flapping[47]. Apple requires the Mac to have been unlocked by another method after the Watch was put on, and limits watch-based iPhone unlock to 6.5 hours since a real unlock[1]. Enterprise systems are slower still. Microsoft Entra's continuous access evaluation propagates revocation in "up to 15 minutes" with session tokens valid for up to 28 hours[48]. NIST SP 800-63B-4 caps the highest assurance sessions at 12 hours and does not describe continuous biometric authentication at all[49].

These windows matter because the people best placed to exploit them are close. In a SOUPS 2016 study, an estimated 20% of US adults had looked through someone else's phone in the past year[50]. On the lab side, coverage of the Linwei Ding indictment reported an allegation that a colleague scanned Ding's badge while he was abroad[51]. Ding was convicted in January 2026 of economic espionage and theft of AI trade secrets[52]. A presence heartbeat narrows the handover window to its renewal interval, but it reports only that *a* matching face is visible. It cannot tell whether the hands on the keyboard belong to that face.

AI agents now operate computers from inside the owner's session

Computer-use agents have moved onto the owner's real desktop. Anthropic's Claude Code computer use "runs on your actual desktop with access to the apps you approve". It needs the macOS Accessibility permission, which "lets Claude click, type, and scroll"[26]. Microsoft's Copilot Actions is "an AI agent that completes tasks for you by interacting with your apps and files ... to click, type, and scroll like a human would". Microsoft names cross-prompt injection, which can cause "data exfiltration or malware installation", as a new risk[27]. OpenAI's Operator and Google's Gemini computer-use model act mainly from browsers the vendors run[53, 54]. CAPTCHAs no longer separate these agents from people: an ETH Zurich team solved 100% of reCAPTCHAv2 challenges with off-the-shelf vision models[55].

This changes what a gate can promise. An owner who launches an agent and then sits at the desk passes every presence check, while the agent, or a prompt-injected agent, or a remote-access trojan using the same synthetic-input permission, acts with the owner's full authority. Not-a-Bot, the best-known research system for attesting human input, states the structural limit plainly. Attesting that real input occurred recently "does not prevent bots from piggybacking on human activity"; it only caps their rate[56].

Injected input looks like hardware

Operating systems can flag input injected through their user-mode APIs. Windows marks hooked keystrokes with `LLKHF_INJECTED`, and macOS gates synthetic events behind a separate PostEvent permission[57, 58]. These flags describe a software path, not a human. Since Windows 10, a kernel-mode "HID source driver" can create virtual devices whose reports the HID class driver enumerates "similar to how it enumerates those collections for a real HID device", and the driver sets its own vendor ID, product ID and container ID[24]. The O.MG Cable is sold as a USB cable with a "WiFi implant inside", offering keystroke injection and a hardware keylogger (vendor claim)[25]. The Rubber Ducky is reported to type up to about 890 characters per second with randomized jitter to evade detection (secondary)[59]. Even protected prompts have fallen. Patrick Wardle showed that macOS once converted a synthetic mouse event into one with source PID 0, which security prompts then accepted as trusted[60]. On Windows, the Secure Attention Sequence itself can be generated in software when policy allows[61].

Real-time deepfakes and camera injection defeat face presence on an unattested path

Injection, not masks, is the dominant attack on camera biometrics. iProov's threat report counts native virtual-camera attacks up 2,665% in 2024 (vendor data)[62]. CyberArk unlocked Windows Hello by feeding a single valid infrared frame from a custom USB device (CVE-2021-34466)[63]. Researchers earlier fooled it with printed near-infrared photos, even with enhanced anti-spoofing enabled[64]. At Black Hat 2025, ERNW showed that a local administrator could insert face data into the Windows Hello database so that any face unlocked the PC. The hypervisor-isolated Enhanced Sign-in Security (ESS) mode blocks that attack[32]. Pulse-based liveness no longer separates real faces from synthetic ones. Real-time face swaps inherit the driving person's heartbeat, so a deepfake shows a plausible pulse[16]. The financial stakes are concrete. Engineering firm Arup reportedly lost about US$25 million after an employee joined a video call in which every colleague was synthetic (secondary)[65].

Relay attacks turn proximity into distance

NCC Group demonstrated a link-layer relay attack on Bluetooth Low Energy in 2022. It convinced devices that a key was nearby "from hundreds of miles away", defeated encryption and latency checks, and affected phones, laptops, smart locks and cars[66]. Google warns that "location signals can be copied or manipulated"[2]. Apple's Auto Unlock uses peer-to-peer Wi-Fi time-of-flight ranging rather than signal strength. A 2017 TU Darmstadt thesis nevertheless reported a man-in-the-middle attack on it, and the thesis page does not say whether Apple fixed the issue[1, 67]. A gate that infers presence from radio proximity inherits all of this.

Administrator compromise removes any enforcement that lives on the endpoint

SentinelOne's incident data show security tools disabled in about 48% of 2024 intrusions, and "EDR killers" are now bundled into ransomware-as-a-service[31]. Any verification agent that runs as ordinary software on a commodity OS shares a privilege domain with the attacker it is meant to stop. It can be killed, fed injected frames, or have its template store edited[32].

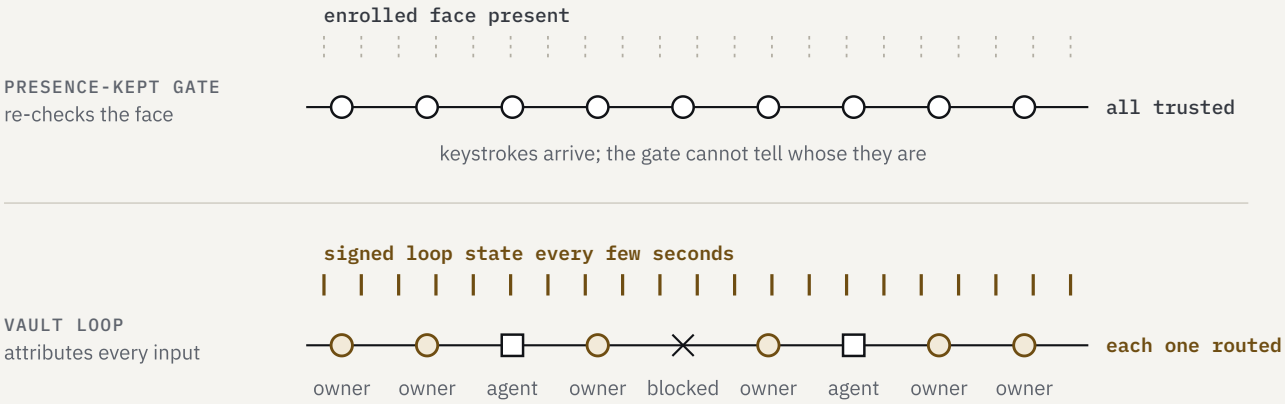
Coercion succeeds against any loop as well as any gate

RAND's weight-security framework scores "extortion" at 4 of 5 (a 60–80% chance of success) for organized-crime and standard state attackers. It notes that "leverage can also be applied to friends and family of the target", and its only top-tier countermeasure is protecting staff physically[68]. Consumer coercion is now measurable. CertiK counted 72 verified crypto wrench attacks in 2025, up 75%, with more than $40.9 million in confirmed losses. Its early-2026 count was running 41% ahead, with victims' family members increasingly targeted[41, 69]. Jameson Lopp's catalogue of about 225 verified attacks finds about 66% succeed, and nearly a quarter are home invasions (secondary)[70]. Coercion is the one threat that defeats the loop as completely as the gate. A coerced owner is a living, enrolled body producing genuine input. Section 9 explains why the answer is procedural.

The threat matrix: five of eight threats run with the owner present

THREAT	CONSUMER INSTANCE	LAB INSTANCE	WHY A PRESENCE-KEPT GATE PASSES IT	WHAT THE LOOP MUST ESTABLISH (INFERENCE)
Handover after login	Family member or thief at an unlocked laptop during the relock window[2, 50]	Badge or session lent to a colleague (alleged in the Ding case)[51]	Presence is renewed by any matching face or nearby token; hands are never checked	P2 carried by custody plus P3 per input; a contradiction ends authority at once
AI agent acting in the owner's session	Desktop agent or prompt-injected agent driving apps[26, 27]	Coding or operations agent with access to weight infrastructure	Owner is present and verified while the agent types	P3 with three-way attribution; agents as separate principals
Malware piggybacking on real activity	Remote-access trojan using the PostEvent or SendInput path	Same, after endpoint compromise	Recent genuine input does not prove origin[56]	Per-event binding of each input to the body
Injected HID device	O.MG cable, Rubber Ducky[25, 59]	Same, or kernel virtual-HID driver[24]	The OS sees "hardware"	Attested input path; unknown devices cannot carry P3
Real-time deepfake or camera injection	Virtual camera feeding the owner's face[62]	IR-frame injection[63]	Face presence on an unattested path is forgeable	Attested camera path; cross-check with on-body channels the attacker does not hold
Proximity relay	BLE relay of the owner's watch[66]	Same, for badge or band	Proximity is inferred from radio presence	Secure ranging plus custody; radio presence never counts as identity
OS or administrator compromise	Infostealer with admin rights	EDR killer, template injection[31, 32]	Enforcement lives on the compromised endpoint	Enforcement point outside the OS; fail closed on missing loop state
Coercion	Wrench attack at home[41]	Extortion of operator or family[68]	Everything is genuine	P4 cannot be measured: covert signal, quorum, delay, silent degradation

Five of these eight threats can proceed while the verified owner is present. That is the core reason the proposition "an enrolled person is here" is the wrong thing to keep alive. The loop has to prove that the inputs are the owner's, and it has to be designed so that an owner who is present but not free cannot deliver the payoff alone.



Gate vs. loop: the question kept alive. A gate that re-checks "is the enrolled face still here?" trusts every keystroke inside the session alike: the owner's, an AI agent's, or an injected device's. The loop asks a question per input instead. Body-bound owner input reaches the vaulted context. Agent input goes to the agent's own workspace under an owner-signed mandate. Unattributed input is blocked and counted as a contradiction[24, 26, 56].

What the loop must prove, and the shape of the machine that proves it

Four propositions, three of them measurable

The loop's invariant breaks into four propositions with very different evidence behind them. The table states what each one asks, which evidence can establish it, and what the loop does when it fails. The responses are this report's proposals.

PROPOSITION	QUESTION	EVIDENCE THAT CAN ESTABLISH IT	EVIDENCE THAT CANNOT	RESPONSE WHEN IT FAILS (PROPOSAL)
P1: a living body is present and attached	Is there a living human, and is it the one wearing the loop's sensors?	Contact pulse (ECG, PPG), interstitial chemistry, EMG, radar breathing, coherent cross-sensor timing[7, 71]	Implants, which prove possession of a key but not life (inference)[72]; any single periodic signal, which can be replayed[73]	Missing evidence: decay trust and re-anchor. Contradiction: end authority
P2: it is the enrolled person	Is this body the owner (or a named successor)?	Hard biometric on an attested path at anchor events (IR face under ESS, Touch ID, iris), carried forward by unbroken custody[15, 74, 75]	Pulse agreement, behaviour alone, possession of a token	A new or mismatched face ends authority at once; a custody break forces re-identification
P3: the inputs come from that body	Was this keystroke, click or command produced by the attested body?	Attested input path plus per-event body binding (EMG or wrist motion preceding each input)[76, 77]	Operating-system injection flags against an administrator[24]; window-level correlation[18]	Unattributed input is blocked; agent input is routed to the agent principal
P4: the person is acting freely	Is the owner coerced?	Only an explicit covert signal chosen by the owner[38]	Stress physiology, voice stress, micro-expressions[35, 78]	Never inferred; designed around with quorum, delay and indistinguishable degradation

The first three propositions are measurable, though at very different maturity. The fourth is not, and the National Academies' review of the polygraph is a durable warning against pretending otherwise. It found no alternative physiological or behavioural method that "has yet been shown to outperform the polygraph", and judged polygraph screening "an unacceptable choice" for employee security screening[78]. The loop therefore treats freedom as a design constraint, not a measurement. It never claims to detect coercion. It makes coercion unprofitable and gives the owner a covert way to say it is happening.

The architecture: channels, cross-checks, a trust engine and an enforcement point

The loop has four functional layers and one recurring ceremony (proposal, assembled from the evidence in Sections 4–8).

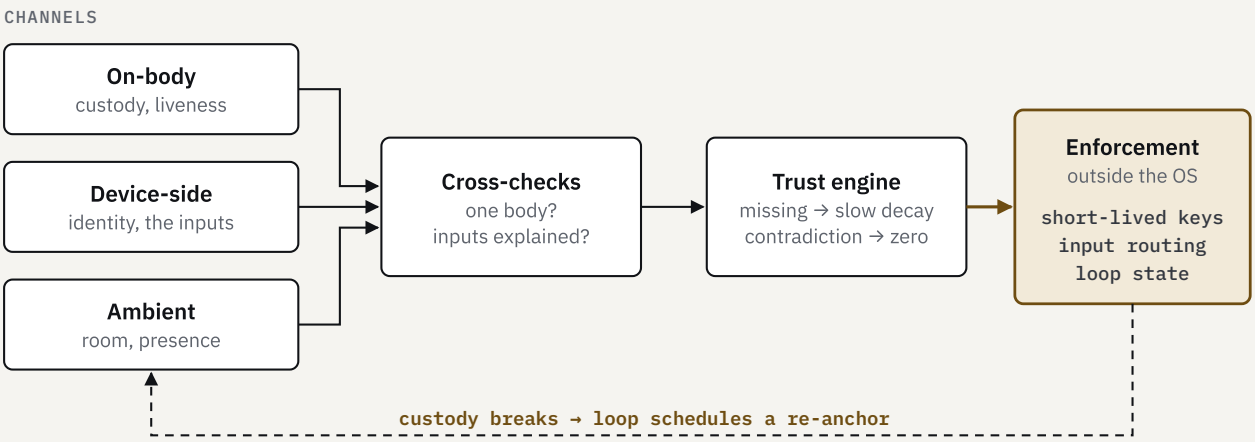
Channels produce raw evidence from three places. On-body carriers include a wristband, ring, patch, microneedle sensor or earbud, which provide liveness and custody and sometimes a weak identity signal. Device-side sensors include an attested IR camera, a fingerprint sensor in the keyboard and the input devices themselves, which provide identity and input events. Ambient sensors include radar presence, the webcam's pulse signal and second-person detection, which report the room. Every channel must authenticate its own output to the enforcement point, following the Magic Keyboard pattern. In that pattern the sensor holds an attestation key, pairs to the host's secure element through authenticated key exchange, and encrypts its traffic, so the host knows a reading came from a genuine sensor[15].

Cross-checks compare channels against each other. The same heartbeat should appear at the wrist, in the face video and in the radar return, with physiologically fixed delays. Each keystroke should be preceded by matching wrist motion or muscle activity. The wrist should stop typing when the keyboard stops. The design rules for these checks come from the attacks on earlier systems and are set out in Section 5.

The trust engine fuses evidence over time into a running estimate that the invariant holds. It distinguishes *missing* evidence, which decays trust slowly toward a floor, from *contradictory* evidence, which cuts trust to zero. It groups channels into failure domains so that one cause, such as bad lighting, counts once. It requires every core role to be covered, with grace windows for gaps. Section 7 gives the arithmetic.

The enforcement point converts trust into capability. It releases short-lived keys, routes or blocks each input, and moves the system between normal, degraded, re-anchoring and duress-shadow states. It sits outside the general-purpose OS. On the lab tier that means the terminal's tamper-responsive secure element streaming signed loop state to a confidential-computing keyserver. On the consumer tier it means a secure element, either the platform's own for per-action key release or a Vault hub (Section 8).

Anchoring is the recurring ceremony that connects identity to custody. A strong identity check on an attested sensor binds a freshly attached on-body carrier to the enrolled person, and custody then carries that binding forward. Anchoring looks like a gate, but its role in the loop is different. It is not a door to a session. It is a state transition the loop schedules for itself whenever custody breaks: a band comes off for charging, a patch is replaced, or a contradiction resets the chain. The shipping precedents already work this way. Nymi says workers authenticate about once per shift with fingerprint plus ECG, and trust is then maintained through continuous on-body detection (vendor claim, sponsored content)[9]. Apple's Watch must be unlocked on the wrist before it can unlock anything[1].



The loop (proposal). Channels on the body, the device and in the room send signed evidence. Cross-checks bind that evidence to one body and ask whether every input is explained by it. The trust engine lets missing evidence decay slowly but cuts trust to zero on any contradiction. An enforcement point outside the operating system turns the result into short-lived keys, per-input routing and loop state. A strong identity check is not a door at the start. It is a step the loop schedules for itself whenever custody breaks[15, 34].

The loop runs at several cadences at once

A loop that tried to establish everything at one rate would be either too slow to catch an injected keystroke or too demanding to wear. The proposal below assigns each proposition its own cadence, and each value is a design target, not a measured result.

CADENCE (PROPOSAL)	WHAT IS CHECKED	TYPICAL EVIDENCE	GROUNDING
Per input (milliseconds)	P3: each keystroke, click or command is attributed to owner, agent or nobody	Attested input report plus wrist motion or EMG event; OS path and process identity	NAB attests within about 1 s of input[56]; per-event matching is needed against piggybacking (inference)
Per epoch (1–5 s)	P1 and custody: on-body carriers report tissue contact and pulse; signed loop state renews the session key	PPG or ECG, skin contact, secure ranging	RATS freshness and watchdog pattern[34]; Apple's ranged unlock[1]
Per minute	P2 refresh: passive face match on the attested camera when the face is visible; cross-modal pulse coherence	ESS IR face; rPPG versus wearable pulse	Hard biometrics reserved for re-authentication, soft cues for tracking[79]
Per custody break (daily)	Re-anchor after band removal, charging or a contradiction	Touch ID, ESS face, ECG at don	Nymi and Apple Watch pattern[9, 80]
Per patch (5–15 days)	Apply a new adhesive or microneedle seal under a strong identity check; refresh physiological templates	Patch application ceremony	Wear limits of patches and CGMs[81, 82, 83]

CADENCE (PROPOSAL)	WHAT IS CHECKED	TYPICAL EVIDENCE	GROUNDING
Per year or event	Re-enrol templates; review heirs, agents and quorum	Supervised ceremony	Template drift[13, 84]; succession patterns[85]

What changes relative to a presence-kept gate

Relative to a presence-kept gate, six things change. Authority shrinks from a session to an input and a few-second epoch, so a lent badge or a stolen token is worth seconds, not hours. The proposition kept alive becomes "these inputs are explained by the owner's attested body", which is what defeats agents posing as the owner, piggybacking malware and handover. Contradictions act at once while missing evidence decays slowly, which spares the idle owner but not the impostor. Re-anchoring becomes a state the loop schedules for itself, not a door the owner must walk back through. Agents become first-class principals instead of invisible passengers in the owner's session. And duress becomes a loop mode: a shadow state that looks normal to an observer.

Liveness is plentiful, identity is concentrated, custody is a negative signal

Each candidate channel is judged here on four separate properties. Liveness (P1) asks whether it proves a living human is attached. Identity (P2) asks whether it proves *which* human. Custody asks whether it proves the same body has stayed attached since the last strong check. Input binding (P3) asks whether it can tie individual inputs to the body. Few channels score well on more than one property, which is why the loop needs several channels at once. The reason is not that many weak identity signals add up to a strong one, which Section 7 shows they often do not. The reason is that each proposition needs its own kind of evidence.

On-body channels carry liveness and custody, and rarely identity

ECG is the most studied cardiac biometric, and shipping products have moved away from it as an identity anchor. The original Nymi Band identified wearers from the shape of the ECG waveform and stayed authenticated only while worn[86, 87]. Halifax's 2015 bank trial added separate sensors to confirm the wearer had not removed it[88]. Nymi Band 3.0 paired a fingerprint sensor (vendor-claimed false positive rate about 1 in 500,000) with an ECG check that the finger and the wrist belong to one body[89]. The current product page lists only a fingerprint sensor, FIDO2 certification, "continuous authentication" and a battery life of "3+ days" (vendor)[90]. Cross-session drift explains the shift. EDITH's identification fell below 80% on CYBHi, whose sessions are about three months apart[84]. A September 2026 study's 1.7% equal error rate (EER) from rest to peak exercise was measured within one session, and its authors warn that such protocols can confuse identity with electrode placement and posture[91]. Electrodes also accept injected signals. *Broken Hearted* (NDSS 2017) fooled the Nymi Band in 81% of attempts with Nymi-recorded ECG, 62% with mapped ECG from other devices and 43% with raw ECG from other devices, delivered by waveform generators, a sound card or `.wav` files played from an audio player[14]. Requiring heart-rate variability and a simultaneous PPG signal to agree with the ECG defeated these attacks only in the authors' simulations[92]. Apple Watch's ECG app needs a deliberate 30-second finger touch and has no identity function[93]. Prescription patches record continuously for 7–14 days (Zio field wear averages 12.6 days) and have no identity function either[81, 94, 95]. ECG therefore works best at don time, as a same-body check ("this finger and this wrist share one heart"), and then as a waveform that must stay coherent with a second cardiac channel (inference).

PPG (optical pulse) is the most available continuous signal and the weakest identifier. The best real-world evaluation tracked 44 students over 13 weeks. Methods that scored well on lab datasets collapsed: one fell from MCC 0.94 to 0.44, another from 0.70 to 0.09. Training on the first half of the semester and testing on the second dropped identification to MCC 0.07–0.08[13]. NDSS 2026's wrist-PPG continuous-authentication study reports 88.11% accuracy, a 2.76% EER and an 11.77% false-reject rate across 26 subjects. Its models were specific to each physical device, so a replacement watch means re-enrollment, and spoofing was not evaluated[96]. No study tested identification from a smart ring alone. The closest work verifies ring or watch PPG against the face's camera-measured pulse, at 95.5% balanced accuracy with 33 subjects, which binds a wearable to a face rather than identifying a person[97]. PPG's value is liveness and removal: a pulse that disappears means no tissue. Its weakness is people. Apple confirmed in 2015 that dense dark wrist tattoos can interfere with the heart-rate sensor, enough for the watch to think it is off-wrist and lock[98].

On-body custody is the shipping pattern, and its transfer gap is the attack surface. Apple Watch "locks automatically soon after it's removed from the user's wrist" when wrist detection is on, and device management can enforce the setting[80]. Auto Unlock connects over Bluetooth LE and then uses peer-to-peer Wi-Fi to "securely approximate the distance", sending its unlock secret through a mutually authenticated protocol that terminates in both Secure Enclaves[1]. Android's accelerometer-only on-body detection is the cautionary case: it cannot recognize who holds the phone, so a handed-over phone stays unlocked[44]. A US patent on persistent authentication compares optical and capacitive sensor signals and notes that redundant sensors make a wearable "more difficult to trick"[99]. This research infers that transfer resistance

depends on three things: how many independent continuity sensors there are, whether one of them carries identity (an ECG or pulse-shape template that breaks coherence on a swap even if skin contact is kept), and how fast the session dies after a break. No source describes a wearable that *cryptographically proves* it never left the body. Custody is always a negative signal: no break detected.

Surface EMG is the only on-body channel that could plausibly serve liveness, identity and input binding at once, and it is locked inside one ecosystem. Meta's *Nature* 2025 paper trained generic decoders that work across people from a "highly sensitive, easily donned sEMG wristband" and thousands of participants. It reports a median 0.88 gesture detections per second and 20.9 words per minute of handwriting, with personalization improving handwriting by 16%[100]. The Meta Neural Band shipped with Ray-Ban Display glasses at $799 in the US from 30 September 2025. Telekom has sold the bundle in Germany since 13 October 2026, and the band is not sold separately[101, 102]. As a biometric, wrist EMG reached a median multi-day EER of 2.5% on GRABMyo (43 participants, 3 days)[103], and 1.1% with a user-customized gesture code[104]. A review cautions that most EMG-biometric studies have fewer than 25 subjects, the larger ones use a single day, and electrode shift, sweat and exercise degrade consistency[105]. Forearm EMG fires before a keystroke, and an observer cannot see it to imitate it. That makes EMG the most promising signal for binding input to the wearer. **No paper validates EMG-to-keystroke attribution as a security primitive** (inference and gap). This research found no public API that lets a PC pair with the Neural Band (gap).

Ear and head channels offer passive identity in narrow form factors. NEC's ear-canal acoustic authentication measures resonance in about one second and claims above 99% accuracy (vendor claim). Its Hearable Device went on sale in Japan in 2020[106, 107]. University at Buffalo's EarEcho reached about 95% at one second and 97.5% with continuous monitoring, but on 20 subjects[108]. Apple Vision Pro shows the target behaviour in a headset. Optic ID claims a random-person false match below 1 in 1,000,000 (vendor claim), and removing the headset locks it[75, 109]. Apple does not document whether Optic ID re-checks during a session. Meta's Ray-Ban Display glasses have no eye tracking and so cannot contribute iris identity[110].

Implants are possession tokens under the skin. VivoKey Apex is an NFC secure element running a FIDO2 applet, vendor-claimed to Common Criteria EAL6+[72]. It proves possession of a key at centimetre range. It does not prove liveness, since a chip answers just as well from a severed hand or a cadaver, and it cannot be read continuously from a laptop (inference). Washington's HB 2303 (2026) bars employers from requiring microchip implants and gives workers a right to sue, and at least a dozen states have similar laws[111, 112]. Implants can only ever be an optional extra factor for either tier.

The blood-sample verdict: no continuous identity, but the strongest custody seal

The founder's suggestion of "a blood sample" deserves a precise answer, because it contains one idea that fails and one that is better than it looks.

As **continuous identity**, blood fails on every axis the loop needs. DNA is definitive, but the FBI-approved Rapid DNA booking system takes about 90 minutes to process, enroll and search a sample (vendor release)[6]. DNA is also not a secret. In a 2024 University of New Hampshire study, one person's DNA transferred to a second person's hand in 50% of trials[21], so holding a matching sample proves little. For relatives, DNA is probabilistic. Even at 40 STR loci, half-siblings showed 5–7% false inclusion or exclusion[113]. Biochemical signatures are only partly individual. In the KORA cohort, metabolic profiles uniquely identified just over 40% of 818 participants after seven years[22]. Infrared fingerprints of serum and plasma were "highly stable" within individuals over six months, but in a 31-person study using laboratory instruments[114]. A 2021 conference abstract claimed individuals could be recognized from continuous glucose traces, but its accuracy could not be verified[115]. No glucose stream identifies a person in real time.

As a **custody and liveness seal**, a minimally invasive sensor is the best on-body channel available. The FDA's decision summary for Biolinq Shine, the first autonomous microneedle glucose sensor (De Novo, September 2025), says the device is worn on the inner forearm only, is single-use, and "cannot be re-applied if it falls off"[7, 116]. First-generation wear is battery-limited to about five days[83]. Filament CGMs run 15 days. Dexcom's G7 15 Day is indicated for 15.5 days, but only 73.9% of sensors lasted the full period in Dexcom's study, and each needs a 60-minute warm-up[82, 117]. A sensor that sits partly inside the body, breaks physically on removal, cannot be re-applied, and streams interstitial chemistry that a dead or synthetic surface would not produce is far harder to move to another body than a clasped band. This research infers a design. Apply the sensor during a strong identity ceremony, pair it cryptographically to Vault, and treat the session as valid only while it reports physiologically plausible data. That makes a 5–15-day custody seal. It proves P1 and custody, never P2.

Three caveats bound the idea. First, no CGM maker documents a security-grade removal event. Removing a Dexcom sensor is described only as peeling it off "like a bandage", which ends the session[23]. Removal latency is unknown, and no study has tried spoofing the sensor's "living tissue" signal, for example in a glucose solution. Vault would need vendor cooperation or custom hardware. Second, adhesives cause allergy. In a 2024 Swedish study, about 63% of 54 diabetes patients referred for suspected device dermatitis were allergic to isobornyl acrylate, against 0.2% of general dermatitis patients. That is a referred population, not a prevalence figure[118]. Third, the cadence is a burden. Reapplying every 5–15 days, plus early failures, means roughly 25–50 identity ceremonies a year (inference). Glucose is also health data, which Section 11 shows carries its own legal weight. The verdict is to offer the microneedle seal as an optional high-assurance custody carrier, for example for a crypto signing session or a lab operator's shift. It should never be a required channel. A one-off lab draw at enrollment or succession adds little that a supervised in-person ceremony with documents and fresh biometric enrollment does not already provide. Sections 10 and 11 explain why DNA should not gate heirs.

Device-side channels supply identity and the input events themselves

Infrared face on an attested path is the strongest shipping passive identity channel on a laptop, and it exists only on Windows. With Enhanced Sign-in Security (ESS), Windows runs face matching in virtualization-based security (VBS). The hypervisor lets only the face camera write into protected memory, templates are encrypted under keys only VBS holds, and a successful match authorizes Windows Hello keys through a secure channel to the TPM[74]. ESS is on by default on all Copilot+ PCs. It "isn't supported for external camera modules", and on multi-user devices the lowest Windows Hello posture applies[74, 119]. Macs had no Face ID as of late 2025, when Bloomberg reported it "remains years away" (secondary)[120]. Continuous face presence on a Mac would therefore be third-party software without an isolated camera path. Microsoft does not document whether ESS can re-verify continuously rather than only at sign-in (gap).

Fingerprint in the keyboard is the reference design for an attested peripheral. The Magic Keyboard with Touch ID stores no templates. Matching happens in the Mac's Secure Enclave, the keyboard attests with a hardware public-key accelerator, and Touch ID data travel AES-GCM encrypted over ephemeral ECDH[15]. Apple makes these claims only for Touch ID data, not for ordinary keystrokes, and a read needs an explicit touch. Fingerprint sensing under every typing key exists only in a patent[121]. A 2005 research system put a fingerprint reader in the mouse and fused it with camera face verification over time[122]. Fujitsu's palm-vein mouse needs the palm held about two inches above the sensor, so it suits a challenge, not continuity[123].

Behaviour and gaze are passive but slow and forgeable. Mondal and Bours' trust model needed an average of about 275 actions to detect an impostor[124]. Conditional generative models now synthesize keystroke dynamics that fooled an external authentication system[125]. Eye-movement biometrics stayed reliable over two weeks against "lunchtime" attackers[126]. Their errors concentrate on particular users, so a good mean EER can hide a few owners who are locked out constantly[127]. Tobii Aware ships eye-tracking presence, auto-lock and a "Continuous ID" refocus feature on Lenovo, Acer and MSI machines (vendor)[128]. These signals belong in the loop as low-weight corroboration. They also carry legal weight, because Washington counts keystroke rhythms as biometric health data (Section 11).

Ambient channels report the room, not the person

Webcam pulse (rPPG) is a conditional channel. Under controlled lighting, classical methods estimate heart rate within about 2 BPM on the standard benchmark[129]. Models tested on the mobile, mixed-skin-tone, mixed-lighting MMPD dataset err by 9–18 BPM[130, 131]. Intel claims 96% accuracy for its rPPG-based FakeCatcher (vendor claim). An informal BBC test reportedly found it flagged several real videos as fake (secondary)[132, 133]. Its role is a cross-check, never a channel that must always pass.

Millimetre-wave radar ships for presence. Google's Nest Hub senses breathing with Soli radar[134], and Infineon markets 60 GHz presence parts for notebooks, PCs and monitors (vendor)[135]. Vital signs need stillness. In TI's user forum, staff said the person must stay still for at least 30 seconds facing the antenna (user forum material)[136]. Radar identification is lab-only. Cardiac Scan reported 98.61% balanced accuracy and a 4.42% EER on 78 subjects[137]. Radar identification is also spoofable: researchers synthesized cardio-respiratory signatures with a phase shifter driven by a signal generator, and replayed recorded chest motion through an actuator hidden under clothing[73, 138].

Presence and second-person detection complete the ambient group. Windows presence sensing targets under 5% false wakes and under 5% missed wakes. It is unavailable on some machines with the lid closed or an external display attached[47, 139]. Commercial continuous-face agents lock on "additional or unknown faces in the frame" (vendor)[11]. Presence must never count as identity. It is a trigger: start a grace timer when presence drops, and demand re-verification when it returns.

Scorecard: three channels carry hard identity, none carries input binding

The matrix below is this report's synthesis of the evidence above (inference, not a measured benchmark). "Strong", "moderate" and "weak" compare channels with each other for the stated property. They are not certified ratings.

CHANNEL	LIVE BODY (P1)	IDENTITY (P2)	CUSTODY	INPUT BINDING (P3)	STRONGEST ATTACK EVIDENCE	CONSUMER AVAILABILITY	LOOP ROLE (PROPOSAL)
Wrist ECG band (Nymi-style)	Strong	Moderate at don; drifts across sessions[84]	Needs separate on-body sensor	None	Injection 43–81%[14]	Enterprise only	Don-time same-body check; cardiac coherence
Chest ECG patch	Strong	None offered	Adhesive seal, 7–14 days[81]	None	Unstudied	Prescription only	Lab-tier custody seal option
Watch or ring PPG	Strong	Weak in the field[13]	Good (tissue presence)	Weak (motion)	Optical injection unstudied	Ubiquitous	Core custody carrier; liveness
Wrist detection plus secure ranging (Apple Watch)	Implicit	Inherited from unlock	Locks on removal; 2–3 m ranging[1]	None	MitM on Auto Unlock (2017, status unknown)[67]	Mac only	Core custody carrier on macOS
sEMG band	Strong	About 2.5% EER multi-day (lab)[103]	Good	Promising, unvalidated	Unstudied	Bundled with Meta glasses only	Research channel for P3

CHANNEL	LIVE BODY (P1)	IDENTITY (P2)	CUSTODY	INPUT BINDING (P3)	STRONGEST ATTACK EVIDENCE	CONSUMER AVAILABILITY	LOOP ROLE (PROPOSAL)
Microneedle or CGM sensor	Strong (live interstitial fluid)	None	Single-use, cannot re-apply[7]	None	Unstudied	OTC CGMs; no security API	Optional high-assurance seal
DNA or metabolome	Not applicable	DNA definitive but ~90 min[6]	None	None	DNA is shed[21]	Lab only	Excluded from the loop
NFC or FIDO implant	None	Possession of a key	None (centimetre tap)	None	Passive chips clonable	Biohacker niche	Optional extra factor only
Ear-canal acoustics	Moderate	Above 99% (vendor claim); 95–97.5% on 20 subjects[106, 108]	Good (ear seal)	None	Unstudied	Japan only	Optional passive identity
Headset iris (Optic ID)	Yes	Below 1 in 1M FMR (vendor claim)[75]	Locks on removal	None	Not in sources	Vision Pro	Lab option for headset users
IR face on attested path (ESS)	Moderate (IR)	Strong	Only while face visible	None	IR photo (older builds), USB IR-frame injection[63, 64]	Copilot+ PCs; built-in only	Core identity anchor and refresh (Windows)
RGB webcam face	Weak	Strong matcher, unattested path	Only while visible	None	Virtual cameras, deepfakes[62]	Universal	Telemetry; may only lower trust
Touch ID keyboard or ESS fingerprint	Weak	Strong (vendor-claimed FAR below 1 in 50,000 for Touch ID)[140]	None	One bit per touch	Match-on-chip bus MITM (patched)[141]	Mac, Windows	Core identity anchor; secure-intent gesture
Per-key fingerprint	Weak	Strong in principle	None	Strong in principle	None (patent only)	No	Phase 3 research
Keystroke and mouse dynamics	None	Weak; ~275 actions to flag[124]	None	Weak	GAN-forged keystrokes[125]	Software	Low-weight corroboration
Gaze (eye tracker)	Moderate	Moderate; per-user systematic errors[127]	Only while looking	None	Not evaluated	Tobii Aware laptops	Optional corroboration
Webcam rPPG	Moderate when still	None	None	None	Deepfakes carry the driver's pulse[16]	Universal	Cross-check only, never must-pass
mmWave radar	Moderate (breathing)	Lab-only, spoofable[137, 138]	Seat presence	None	Phase-shifter and actuator replay	About $60 sensor[142]	Presence and timing cross-check

CHANNEL	LIVE BODY (P1)	IDENTITY (P2)	CUSTODY	INPUT BINDING (P3)	STRONGEST ATTACK EVIDENCE	CONSUMER AVAILABILITY	LOOP ROLE (PROPOSAL)
Wrist motion vs input (ZEBRA-style)	Weak	None	None	Moderate if trusted-channel-driven	Opportunistic mimicry ~40%[19]	Watch IMU	P3 corroboration, per event

Three patterns stand out. Liveness is available from almost every on-body and ambient channel. Identity comes almost entirely from three hard biometrics on attested paths (IR face, fingerprint, iris), and only IR face is passive at a desk, only on Windows, and only with a built-in camera. Input binding has no mature channel at all. Those gaps shape the reference builds in Section 8.

Wear burden decides whether a channel stays on

A channel that is not worn proves nothing, and wear compliance is the binding constraint for consumers. In Pew's most recent data (2019), about 21% of US adults regularly wore a smartwatch or fitness tracker, rising to 31% in households earning $75,000 or more[143]. Gartner found that 29–30% of smartwatch and tracker owners had abandoned them[144]. Apple Watch Series 11 is rated at up to 24 hours (vendor claim), and the Neural Band at about 18 hours, so a single wrist carrier drops out of the loop every day while it charges[145, 146]. Accessibility cuts across channels. Industry experience is that about 1–2% of people have unusable fingerprints, and capture quality falls sharply after age 65[147, 148]. Tattoos defeat wrist PPG, and adhesives aggravate itch-prone skin[98, 149]. This research draws the consequence: a consumer Vault needs *equivalent assurance bundles*, for example face plus watch, fingerprint plus ring, or fingerprint plus patch, so that a tattooed, elderly or adhesive-allergic owner can reach the same trust by a different route. A fixed set of required devices would exclude roughly 1–5% of users by construction (inference).

Agreement proves one body, never whose body

What agreement between channels proves

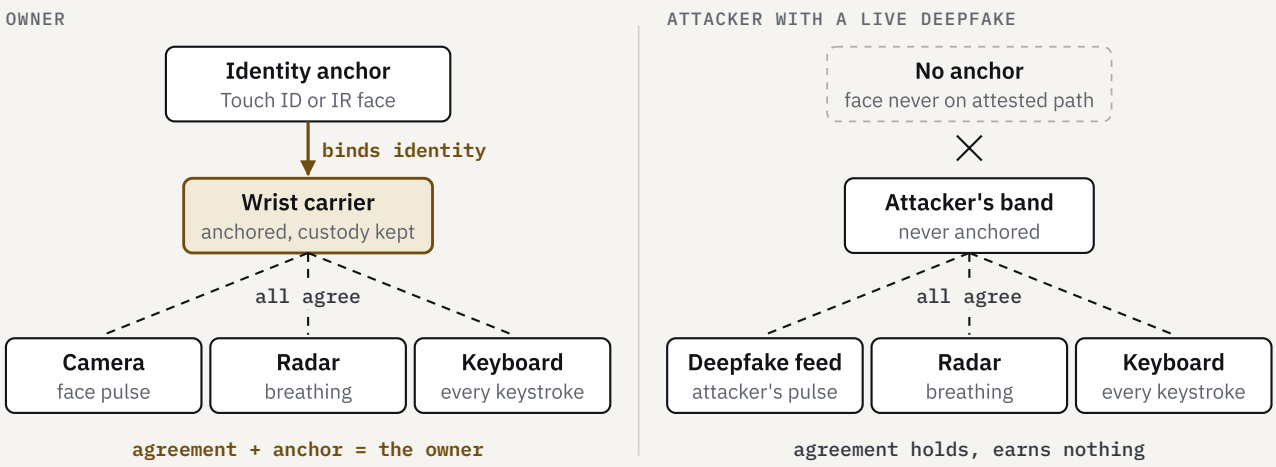
Checking that one physiological event looks the same to two independent sensors is a well-tested security primitive. FaceHeart (INFOCOM 2017) compared the pulse in front-camera face video with the pulse from a fingertip on the rear camera. Against an attacker who replayed the victim's face video on a tablet while covering the rear camera with their own finger, it detected 94.71–98.24% of 340 attack instances. Its EER rose to 8.54% when a gaming task raised heart rate[71]. Heart-to-Heart (CCS 2013) grants a programmer access to an implanted medical device only while both devices record matching ECG, so access lapses the moment physical contact ends[150]. H2B derives a shared key from heartbeat intervals seen by cheap vibration sensors at different body sites, with 95.6% pairing success across 23 participants[151]. Together these show what agreement proves. When independent sensors on the wrist, in the camera and in the radar report the same beat with the right delays, they are almost certainly observing one living body, and that body is in contact with or in view of each sensor.

What agreement does not prove

Agreement does not prove whose body it is. "High-quality deepfakes have a heart!" (Frontiers in Imaging, 2025) found that the heart rate visible in a face swap tracks the heart rate of the person driving it, with r = 0.89 for DeepFaceLive, a real-time tool. The authors conclude that the assumption that deepfakes lack a pulse "is no longer valid". Their own dataset had only 12 people[16]. PulseEdit (IEEE TIFS 2022) edits the pulse in face video without visible change, removing it or inserting a chosen signal, and its authors state it circumvents rPPG-based liveness and deepfake detection[17]. A 2025 method claims to edit physiological signals stably across head motion and skin tones[152]. The consequence for Vault is direct. Suppose an attacker drives a real-time face swap of the owner while wearing a wristband. The webcam pulse and the wrist pulse will agree perfectly, because both are the attacker's heartbeat. rPPG-to-wearable agreement proves that a live heart sits behind the video and that the same body wears the sensor. It does not prove that the face belongs to the owner.

Remote observation also leaks the signals that agreement relies on. Seepers et al. showed that camera-based pulse measurement recovers inter-pulse intervals about as accurately as contact PPG, a "substantial threat" to heartbeat-based security derived from contact PPG. It could not accurately reproduce identifiers derived from electrical ECG signals[153]. Periodic signals can be replayed into sensors, as the radar actuator attack shows[73]. ECG electrodes accept injected waveforms[14]. Every one of these attacks targets a channel whose signal is periodic, observable or physically reachable by the attacker.

The resolution lies in the loop's structure, not in any single check (inference). Identity enters the loop only through a hard biometric on an attested sensor at an anchor event, such as an ESS face match or a Touch ID press. The on-body carrier is attached and bound at that moment. Custody then carries the binding: a band taken from the owner's wrist breaks custody and goes dead, so an attacker cannot wear the owner's anchored carrier. Cross-modal agreement then does the job it can do, which is to bind the face, the inputs and the room to *that anchored body*. In the deepfake scenario above, the attacker's wristband was never anchored to the owner, and the attacker's face never passed the attested camera, so agreement between the attacker's own channels earns nothing. The scenario that remains hard is an attacker who obtains the owner's *live* beat timing remotely and injects it into a forged stream in real time while the owner's anchored carrier is elsewhere. No public work demonstrates that end to end, but its pieces exist[17, 153]. The loop's defence is to require that the face itself arrive over an attested path, and to use timing that a forged stream must match to the millisecond.



Agreement binds; the anchor identifies. When the wrist, the camera, the radar and the keyboard tell one consistent story, they are almost certainly watching one living body. They are not proving whose body it is. A real-time face swap carries the pulse of the attacker driving it, so an attacker's own band agrees with their deepfake perfectly[16, 17]. Identity enters only through a hard biometric on an attested sensor, which anchors the carrier. Custody then carries that anchor forward, and cross-checks bind everything else to the anchored body.

What ZEBRA's failure teaches about binding inputs

ZEBRA (IEEE S&P 2014) is the closest prior attempt at the loop's third proposition. A bracelet on the dominant wrist streamed motion. The terminal classified it into typing, scrolling and keyboard-to-mouse switching, and kept the session only while these matched the actual input events[154]. It verified 85% of legitimate users while identifying all adversaries within 11 seconds, or 90% within 50 seconds at a more lenient threshold[154]. Huhta et al. (NDSS 2016) broke it. Their opportunistic attacker watched the victim at a nearby terminal and mimicked only the interactions they could copy reliably, typically starting to type after the victim started and stopping early. A keyboard-only opportunistic attacker had 70% of windows accepted, and "almost half of the attackers" stayed logged in for the whole experiment. Legitimate users were rejected in only 0–6% of windows, and none was logged out in any 10-minute sample[18]. Aalto's summary reports that opportunistic attackers evaded detection 40% of the time[19]. The root cause, in the authors' words, was "tainted input". The attacker controlled the input sequence and, through its timing, *when* the victim's bracelet data were segmented and compared. ZEBRA also ignored bracelet data whenever the terminal saw no input, a privacy choice, and the authors warn that "pre-mature optimization motivated by privacy ... may introduce security vulnerabilities"[18]. WACA, a later smartwatch-based system, reports about 1% error with 30 seconds of data and robustness to imitation by attackers watching video of the victim. Those are the authors' own figures[155].

Design rules for the loop's cross-checks

The rules below are this report's synthesis. Each comes from a specific attack.

RULE (PROPOSAL)	ATTACK THAT TAUGHT IT	HOW VAULT APPLIES IT
1. The trusted channel drives every comparison	ZEBRA: attacker's input chose when comparisons ran[18]	Wearable and ambient streams run continuously; the verifier asks whether each input is *explained* by them, not whether the wrist matches selected inputs

RULE (PROPOSAL)	ATTACK THAT TAUGHT IT	HOW VAULT APPLIES IT
2. Unexplained activity and unexplained absence both count against the session	Attacker types while the victim uses the mouse; ZEBRA ignored the bracelet when the terminal was idle[18]	Keystrokes without wrist or muscle events, and wrist typing motion without keystrokes, both cut trust
3. Compare every event, at per-event granularity	Selective mimicry; piggybacking on human activity[18, 56]	Each keydown is matched to a wrist or EMG micro-event; no attacker-selectable sampling
4. Attest every sensor, and physically bind it to the body	ECG injection through electrodes[14]; USB IR-frame injection[63]	Sensors pair to the enforcement point with attested keys (Magic Keyboard pattern)[15]; on-body detection guards the skin interface
5. Agreement binds; identity authenticates	Deepfakes inherit the driver's pulse[16]	Identity enters only through a hard biometric on an attested path at anchor events, then rides custody
6. Prefer aperiodic and challenge-driven signals	Heartbeat and breathing replay[73, 153]	Weight keystroke-to-muscle timing and verifier-issued challenges (randomized illumination, LED patterns) over steady rhythms; speech-synchronized signals are harder to fake[156]
7. Keep raw signals for verification but never export them	ZEBRA's privacy shortcut[18]; EMG and motion leak passwords[157, 158]	Process continuously inside the enforcement point's trusted environment; release only verdicts
8. Use fixed physiological lags as anti-forgery timing	FaceHeart lists face-to-finger delay as future work[71]	Require the ECG R-peak to precede the facial pulse by a body-specific delay. This is an untested design idea, and no source gives the tolerance

Rule 8 deserves a caution. The fixed delay between the heart's electrical beat and the arrival of the pulse at the face is a check a forged video would have to reproduce with millisecond timing through any processing pipeline. It is attractive precisely because it is physical. No source measured its stability in a seated user, and no source has tested it against attack, so it belongs in the research plan, not in the v0 design.

Attribute every event to the owner, the owner's agent, or nobody

The third proposition, that the inputs come from the attested body, is the newest requirement and the least mature. It is also the one that separates a loop from a presence-kept gate. It must handle three different sources of non-owner input. Malware and remote attackers want to pass as the owner. Injected devices want to pass as hardware. The owner's own AI agents are legitimate but are not the owner.

Operating systems can flag synthetic input but cannot vouch for a human

Both mainstream desktop platforms record *how* an input event entered the system. On Windows, low-level hooks see `LLKHF_INJECTED` and `LLMHF_INJECTED` flags on events injected through user-mode APIs. `SendInput` is subject to User Interface Privilege Isolation, so a process can inject only into applications at an equal or lower integrity level, and blocked calls fail without any error code saying why[57, 159]. On macOS, posting synthetic events requires a distinct PostEvent privilege that shows up under Accessibility, separate from Input Monitoring[58]. Patrick Wardle reported that macOS 10.14 began ignoring synthetic events globally unless the user grants the app permission; his earlier bypasses had clicked through protected security prompts[60]. These signals describe a software path. An administrator can load a kernel virtual-HID source driver whose reports travel the same class-driver stack as a physical keyboard[24], and a malicious USB device needs no software at all[25]. This research concludes that provenance flags are a **cooperative signal for honest software**, useful for routing input from agents that announce themselves, and not a security boundary against an attacker with admin rights (inference).

The agent question sharpens this. Claude Code's desktop computer use needs Accessibility and Screen Recording on macOS[26]. A remote-access trojan uses the same permission. The operating system can answer "was this event synthetic?" but not "which synthetic source, acting for whom?" (inference). Cloud agents such as Operator act from a remote browser and never generate input on the owner's machine at all[53]. An input-layer loop cannot see them, so their authority has to be governed at the account and service layer.

Research built trusted input paths, and found that input integrity needs output integrity

No shipping consumer platform cryptographically attests ordinary keystrokes. What ships are narrow, hardware-rooted intent and biometric channels. Apple's **secure intent** is "a physical link—from a physical button to the Secure Enclave", designed to confirm a user's intent "without any interaction with the operating system or Application Processor" and to resist software with root or kernel privileges. It ships on Apple-silicon Macs, recent iPhones and iPads, Apple Watch and Vision Pro, and is used for Apple Pay and keyboard pairing[28]. The Magic Keyboard encrypts and attests only Touch ID data[15]. ESS protects only the biometric pipeline[74]. Microsoft's Pluton documentation makes no claims about keyboard or mouse paths (vendor documentation)[160]. The USB-IF Authentication Specification defines certificate-based authentication of USB devices. It authenticates a device's identity, not individual reports, and the research found no evidence of a profile for keyboards and other human-interface devices, or of operating-system adoption[161].

Research systems built the missing path and documented its limits.

SYSTEM	MECHANISM	WHAT IT PROVES	LESSON FOR VAULT
Not-a-Bot (NSDI 2009) [56]	~500-line attester in the Xen hypervisor reads keyboard and mouse directly; TPM-sealed key signs request hash, nonce and time since last input; anonymous attestation	Real input occurred within about 1 s of the request	Bots can still piggyback on human activity at human rates; origin of a specific action is not proven
Bumpy (NDSS 2009) [162]	Keystrokes handled in an isolated Flicker module; user marks sensitive input with `@@`	Marked input bypassed the legacy OS	An 85-person, four-month study found it "requires extra diligence from users"[163]
Fidelius (IEEE S&P 2019)[164]	SGX enclave with a trusted path from input and output peripherals	Secrets protected even if browser and OS are fully attacker-controlled	Trusted path is feasible on commodity CPUs, but depends on enclave security
IntegriKey[76]	External device signs input parameters	Input integrity	Without output integrity, UI manipulation tricks the user into entering the wrong thing
ProtectIOn (NDSS 2020)[76]	Low-TCB device inline between host and IO devices; intercepts HDMI and keyboard and mouse; overlays secure UI	Input and output integrity together, without relying on users checking indicators	The template for an inline Vault input device

Two lessons carry into the loop (inference). Attesting that a human recently typed limits the rate of bot actions but does not prove the origin of any particular action, so the loop needs per-event binding. Protecting input without protecting what the human saw lets malware redirect a genuinely human click, so a confirmation for a consequential action should be rendered by the enforcement point, not by the general-purpose OS. ProtectIOn's inline device satisfies both. It is the closest research analogue to a Vault hub that sits between a desktop's keyboard, mouse and display and the computer.

Body binding turns provenance into a per-event score

Binding each input to the wearer's body converts input provenance from "a keyboard sent this" to "the owner's hand did this". The evidence supports a scored claim, not a cryptographic proof. CSAW, a successor to ZEBRA, uses "temporal features that require any adversary to synchronously mimic the owner"[165]. Venugopalan et al. used forearm EMG as a typing biometric precisely because "keypress timings alone are very easy to spoof"[77]. The same signals leak secrets. *My(o) Armband Leaks Passwords* reached about 76% mean balanced accuracy for keystroke detection, with EMG the most informative channel, over a dataset of more than 310,000 keystrokes[157], and smartwatch motion sensors can infer keystrokes on numeric touchpads[158]. Behaviour alone is not enough either: generative models now forge keystroke dynamics well enough to fool an external authentication system[125].

This research proposes a per-action human-origin attestation of this form: "keystroke *k* at time *t* coincided with a muscle or wrist-motion event in the owner's anchored wearable, with correlation *c*". It is signed at both ends, by an attested input path and by the wearable's own key. Signing both endpoints turns ZEBRA's tainted-input problem into a cross-device consistency check. A fully synthetic stream would then have to forge two hardware-signed sources (inference). The residual attacks are bounded. A nearby human mimic defeats coarse correlation, but Vault's main threat is non-human input, which a mimic does not produce. Malware that times injected input to coincide with the owner's real typing defeats window-level correlation, which is why rule 3 in Section 5 requires per-event matching. And because the signals that make binding possible also leak passwords, raw EMG and motion must stay inside the enforcement point's trusted environment.

Personhood credentials prove something different

The anti-bot ecosystem offers proofs that look relevant and are not.

MECHANISM	WHAT IT PROVES	WHAT IT DOES NOT PROVE
CAPTCHA, Cloudflare Turnstile[55, 166]	A browser session looked human or legitimate at one moment	Anything, against modern solvers: reCAPTCHAv2 was solved 100% of the time[55]
Private Access Tokens (Privacy Pass)[167, 168]	An attested, unmodified device or platform account obtained a token; unlinkable by design	That a human, let alone the owner, produced an input
World ID[169]	The holder has a credential issued to one unique human (iris-verified at an Orb)	That the enrolled human is the one acting now; World's AgentKit deliberately extends it to the human's agents
Personhood credentials[170]	The holder is a real person, without disclosing who	Per-input authorship; the authors argue CAPTCHAs are inadequate against AI

All four are bearer proofs issued at a moment, mostly unlinkable for privacy, with no binding to an input stream. They suit Vault's *outward* face, for example attaching a "verified human principal" claim when an owner-authorized agent acts on the web. They do not serve the local question of who is at the keyboard (inference).

Three principals, and confirmations the agent cannot press

The industry is converging on a pattern Vault can adopt directly. Agents are distinct principals that act under scoped, auditable delegation rooted in a human-signed intent, and they need fresh human presence for consequential steps. Windows 11 gives agents "dedicated agent accounts distinct from the user account" and an "agent workspace" with its own desktop. It requires agents to be "signed by a trusted source so that maliciously or poorly behaved agents can be revoked", and runs Copilot Actions under the agent account to separate "the work done by the agent from other actions on the system"[27, 171]. OAuth Token Exchange records who acts for whom in an `act` claim[30], and an individual draft extending it to AI agents expired in February 2026[172]. Google's AP2 separates a **human-present** flow, in which the user's approval signs a Cart Mandate, from a **human-not-present** flow bounded by an upfront Intent Mandate, creating "a non-repudiable audit trail"[29]. Visa's Trusted Agent Protocol signs agent requests in line with the IETF's Web Bot Auth work, and Mastercard Agent Pay issues per-agent tokens confirmed with payment passkeys[173, 174, 175]. NIST's NCCoE and the Decentralized Identity Foundation both opened agent-identity work in 2026[176, 177].

Mapped onto the loop, every input or action falls into one of three classes (proposal):

PRINCIPAL	HOW THE LOOP RECOGNIZES IT	WHAT IT MAY DO	CONSUMER EXAMPLE	LAB EXAMPLE
Owner-present	Attested input path plus per-event body binding plus anchored custody; P1–P3 hold	Everything the owner's policy allows; consequential steps confirmed by secure intent	Owner signs a wallet transaction with a Touch ID double-press	Operator queries a model inside a confidential VM

PRINCIPAL	HOW THE LOOP RECOGNIZES IT	WHAT IT MAY DO	CONSUMER EXAMPLE	LAB EXAMPLE
Owner's agent	Signed agent binary under a distinct OS account or workspace, holding a delegation token whose root is an owner-signed mandate	Only what the mandate scopes; never vault keys directly; escalates to owner-present confirmation for anything irreversible	Desktop agent drafts email and files in an agent workspace; cannot open the vaulted partition	Operations agent runs evaluations on mock or quarantined weights; cannot trigger egress
Unattributed	Synthetic or HID input that matches neither class, or input with no body binding while the owner's carrier shows idle hands	Nothing: blocked, logged and counted as a contradiction	Rubber Ducky or remote-access trojan typing into a terminal	Kernel virtual-HID driver on a compromised jump host

Confirmation only means something if the agent cannot press the confirm button. Anthropic's guidance asks developers to have a human confirm decisions "that might result in meaningful real-world consequences"[178]. Claude Code's desktop implementation consumes the Escape key "so prompt injection can't use it to dismiss dialogs"[26]. Wardle's macOS bypasses showed why on-screen confirm buttons are not enough: synthetic events once clicked through security prompts[60]. Vault's confirmations for consequential actions should therefore use a hardware secure-intent gesture or an attested biometric match, as Mastercard uses passkeys. The confirmation dialog itself should be rendered by the enforcement point (proposal).

The two tiers apply the model differently. A consumer owner wants agents to work, so the loop's job is containment. The agent runs in its own account and workspace, its synthetic input is routed there, and the vaulted zone (keys, wallets, the password store) never accepts agent input. Each vaulted action needs an owner-present confirmation. At a lab, agents increasingly operate the infrastructure around weights. The agent's workload identity must be distinct from every human operator's. No mandate may grant weight egress. Any agent-initiated step toward export enters the same human quorum and delay as a human request (Sections 8 and 9). For both tiers, the loop should record which principal produced each consequential action in a tamper-evident log, as NIST, the KYA proposals and AP2 all ask[29, 176].

Require every role, give every channel a veto, and let none vouch alone

Combining biometrics can make them worse

The intuition behind "all channels must be on", that more required channels means more security, runs into a classical result. Daugman's analysis of decision-level fusion gives the rules. Under AND, false accepts multiply (FA = FA1·FA2) but false rejects add (FR = FR1 + FR2 − FR1·FR2). His worked example combines a weak test (crossover error 1 in 100) with a strong one (1 in 1,000). The strong test alone makes 200 errors in 200,000 trials, and the AND or OR combination makes 1,100, which is 5.5 times worse. "A strong biometric is better alone than in combination with a weaker one", unless the weaker test's operating point is moved so its false-reject rate falls below twice the strong test's crossover rate[20]. Patent literature calls Boolean AND and OR fusion at fixed thresholds "easily shown to be suboptimal"[179]. The continuous-authentication literature took the alternative route of temporal and score-level fusion. Dynamic Bayesian networks fuse face and mouse-fingerprint evidence across time[122]. Uncertainty-propagating models fuse face, voice and fingerprint scores[180]. Per-action trust models raise and lower trust with every input[124]. Some systems reserve hard biometrics for enrollment and re-authentication and track continuity with soft cues[79].

The arithmetic of a strict AND

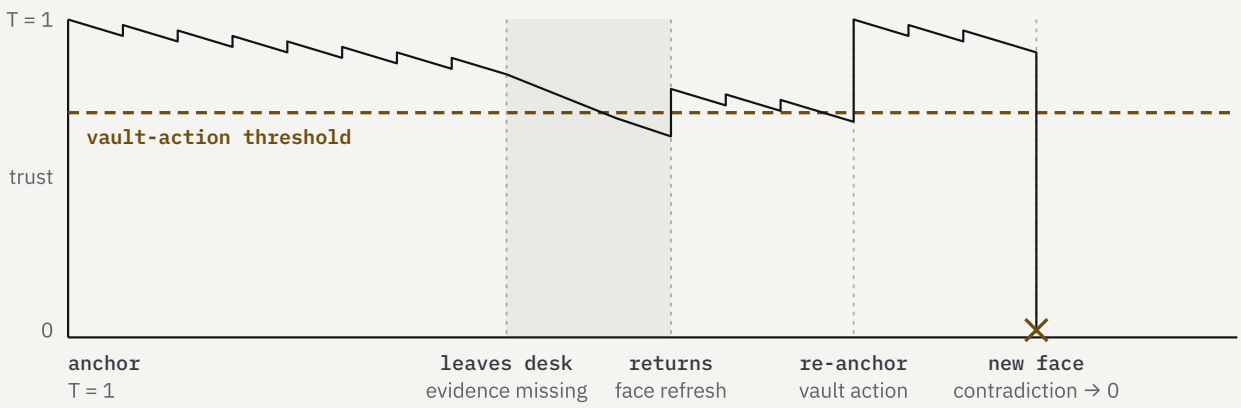
This research works the failure arithmetic for a loop that checks every 10 seconds over an 8-hour day (2,880 checks), assuming independent errors, which is optimistic. With *n* channels each falsely failing with probability *p*, a strict AND fails a check with probability 1 − (1 − p)^n. A grace rule that locks only after *g* consecutive failed checks brings the expected daily lockouts to roughly 2,880 · q^g, where *q* is the per-check failure rate. The extra rows below extend the worked cases using the same formula.

CONFIGURATION (INDEPENDENT ERRORS ASSUMED)	PER-CHECK FALSE-FAIL RATE Q	GRACE G	EXPECTED FALSE LOCKOUTS PER 8-HOUR DAY	ADDED LATENCY BEFORE MISSING EVIDENCE LOCKS
5 channels at 1%, strict AND	4.9%	1	About 141	10 s
5 channels at 1%	4.9%	3	About 0.34 (one every three days)	30 s
5 channels at 1%	4.9%	4	About 0.016 (one every two months)	40 s
5 channels at 2%	9.6%	3	About 2.5	30 s
5 channels at 2%	9.6%	4	About 0.25	40 s
3 core channels at 1%	3.0%	3	About 0.075 (one every two weeks)	30 s

Three conclusions follow. A strict, check-by-check AND across many channels is unusable. Grace windows rescue it at a cost of 30–40 seconds of latency, which sits in the same range as ZEBRA's 11–50-second detection times and Windows' 15-second lock debounce[47, 154]. And the result is very sensitive to each channel's error rate: doubling it from 1% to 2% multiplies lockouts by about seven at g = 3. That is why weak, conditional channels such as webcam pulse or radar heart rate must not be in the AND at all.

Grace windows, failure domains, and the asymmetry that saves the owner

Three refinements make the arithmetic work in practice (inference). **Grace applies only to missing evidence.** A channel with no data, because the owner stopped typing, the room went dark or radar lost the chest, should decay trust slowly toward a floor, as temporal models do. A channel showing *contradictory* evidence should cut trust to zero immediately: a different face, heartbeats that disagree across sensors, keystrokes with no wrist or muscle event, or custody broken on the anchored carrier. This asymmetry defeats the ZEBRA-style attacker while sparing the idle owner, and it means the 30–40-second latency in the table never applies to an impostor swap. **Correlated failures count once.** Face match, face presence and webcam pulse share one camera and its lighting. Radar heart rate and webcam pulse both fail when the owner moves. Keystroke dynamics and wrist-to-input correlation both go quiet when the owner reads. Channels should be grouped into failure domains (camera, RF, worn sensors, input devices), and a shared-cause outage counted as one event. **Error distributions matter more than means.** Eberz et al. showed that eye-movement errors concentrate on specific users, and that optimizing mean EER can reduce security[127]. Vault should measure per-user lockout distributions and report both "time to detect an impostor" and "time to a false lockout", Mondal and Bours' ANIA and ANGA metrics[124].



Missing evidence decays; contradiction ends it (proposal, not a benchmark). An anchor (a hard biometric on an attested sensor) sets trust to 1. Passive refreshes, such as an attested face match, nudge it back up as it decays. Missing evidence, like an owner stepping away, only lets trust fall slowly. Contradictory evidence, like a different face, cuts it to zero at once. A vault action below the threshold makes the loop schedule a re-anchor[124, 181].

Usability targets from the field

Measured long-session lockout data are scarce. ZEBRA's replication found no legitimate user logged out in any 10-minute sample[18]. DARPA's Active Authentication program set the reference target: the system may not "falsely reject the user more than five times in a row during continuous usage over a 40 hour period", and one system let a correct user work a regular day without a false rejection[181]. Microsoft's presence-sensing specification tolerates up to 5% false wakes and 5% missed wakes[47]. In the only user study found, 91% of 37 participants found smartphone implicit authentication

convenient and 35% were annoyed by false rejects[182]. A 2026 preprint proposes a three-zone decision (accept, reject, or inconclusive with re-verification over a shrinking window) and reports faster intrusion detection without hurting genuine users (self-reported)[183]. This research found no published multimodal desktop system reporting false-lockout rates over multi-hour sessions with ambient physiological channels included. Vault's pilots will generate the first such data.

Verdict on "all channels must be on"

Taken literally, the founder's rule means that every channel must pass every check. That version fails three ways: the arithmetic above, Daugman's result that adding weak tests to strong ones can raise total error, and the accessibility evidence that every channel excludes somebody (Section 4). A second reading, that every channel's *hardware* must be powered and attested, fails on charging alone. A daily-charge watch drops out every day, so "watch required" means a predictable lockout.

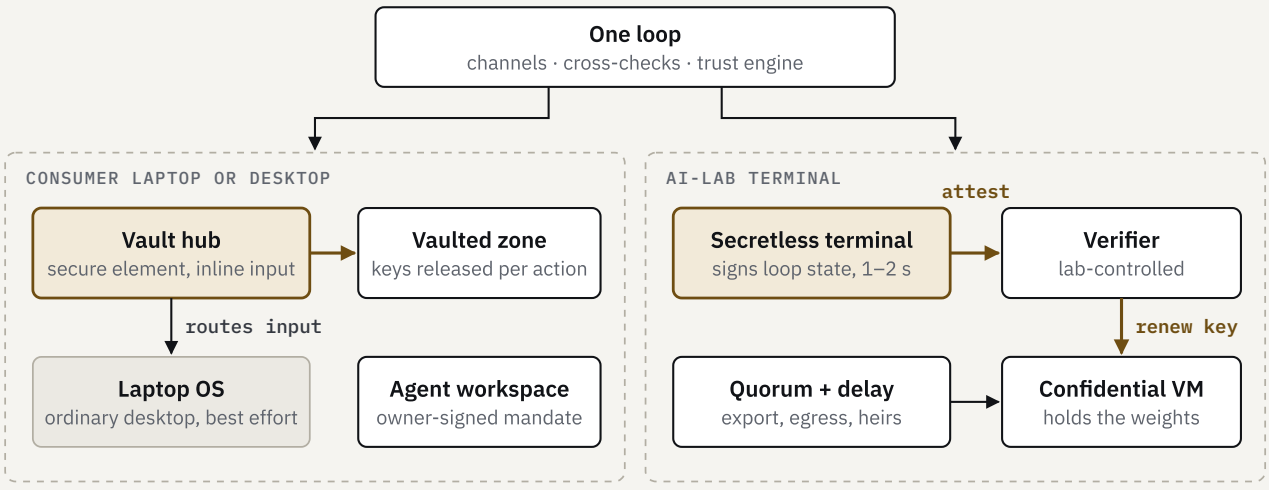
The rule survives in a stronger form that keeps its intent (proposal). **Every proposition must be covered by at least one attested channel at every moment. Every live channel holds a veto. No channel can vouch alone.** Concretely, the loop requires four things at all times. A custody carrier anchored to the owner must be attached, and there should be two overlapping carriers, such as a watch and a ring, or a band and a patch, so that charging one never breaks custody. A hard identity channel must have refreshed within its window. The input path must be attested and body-bound for any input that reaches the vaulted context. No live channel may contradict the others. Weak channels such as webcam pulse, radar, gaze and keystroke dynamics never gate on their own. They shift the operating points of the strong channels, as Daugman recommends, and they can veto by contradiction. A channel that is unavailable to a particular owner (tattooed wrist, worn fingerprints, adhesive allergy) is replaced by an equivalent bundle, not waived. In this form "all channels must be on" becomes the security property the founder intended: no single compromised or absent channel can carry the loop, and any single channel can stop it.

Enforcement and two reference builds

Enforcement has to live where the attacker is not

The loop's verdict is worth only as much as the component that acts on it. Five principles carry across both tiers (proposal, drawn from the threat evidence in Section 2). First, the enforcement point runs outside the general-purpose OS, because endpoint software is routinely disabled[31]. Second, authority is a short-lived key renewed only by fresh, signed loop state, so killing the loop fails closed. This is the IETF RATS watchdog pattern, in which a device lacking "regular and fresh Attestation Results" is forced into a safe state[34]. Third, the enforcement point routes input: it decides per event whether input reaches the vaulted context, the agent workspace, or nowhere. Fourth, confirmations for consequential actions are rendered by the enforcement point and answered by a secure-intent gesture, so neither malware nor an agent can produce them[28, 76]. Fifth, every sensor authenticates to the enforcement point, not to the OS[15, 74].

The signed loop state replaces the presence heartbeat. The proposal is that it carries an epoch identifier; the identity and continuity status of each custody carrier; the time and modality of the last hard-identity refresh; a summary of input-binding results since the previous epoch; any contradiction flags; and the measured boot state. It also carries a duress indicator encrypted so that only the verifier can read it. Its size and timing must not change between normal and duress epochs, so an observer cannot tell them apart (Section 9).



Same loop, two enforcement points (proposal). On a consumer machine, the proposed Vault hub holds the keys to the vaulted zone. It sits inline on the external keyboard, mouse and display and releases keys one action at a time. The rest of the desktop is protected only on a best-effort basis, and a laptop's built-in keyboard can't be put inline. At a lab, the terminal holds no secrets: it streams signed loop state to a verifier, and the confidential VM drops its session key the moment that stream lapses or shows a contradiction[33, 34, 76].

Consumer reference build: commodity sensors, a Vault hub, and an honest boundary

The consumer tier has to run on laptops and desktops people already own. That forces a candid split between what commodity platforms can enforce and what they cannot.

On **macOS**, the anchors are Touch ID (built into Apple-silicon laptops or on a Magic Keyboard) and the Secure Enclave. Secure Enclave keys are used only inside the enclave "after it authorizes their use", and Keychain items can be released "only by a successful match"[184]. The custody carrier is an Apple Watch. It locks on removal, and Auto Unlock ranges the Mac within 2–3 m over peer-to-peer Wi-Fi[1, 80]. Series 11 includes the single-lead ECG that could serve a don-time same-body check, while the $249 SE 3 lacks ECG[185, 186]. Apple's ECG app has no identity function, and third-party access to the trace was not verified[93]. Two gaps remain. Macs have no Face ID, so there is no attested passive identity refresh between Touch ID presses[120]. And this research found no documented auto-lock when the Watch leaves, so Vault would add that behaviour (inference)[1].

On **Windows**, a Copilot+ laptop gives the strongest commodity identity channel. ESS face matching runs in VBS with a protected camera path, on by default, but only on built-in cameras. ESS also supports external match-on-sensor fingerprint readers[74, 119]. Windows presence sensing provides wake on approach and lock on leave on supported hardware[47]. Windows has no first-party equivalent of the Watch's ranged custody protocol. Dynamic Lock only locks, within about a minute, over Bluetooth[46]. A Windows build therefore needs a Vault-provided custody link with secure ranging, not Bluetooth presence (inference).

Commodity platforms leave three things unsolved: a ranged custody link on Windows, an attested input path, and key custody outside the OS. This report proposes a **Vault hub** for all three: a small device with its own secure element. It terminates the wearable links, so custody carriers pair to the hub rather than to the OS. It sits inline between an external keyboard, mouse and display and the computer, in the manner of ProtectIOn[76]. It signs input reports, renders confirmations, and holds the keys to the **vaulted zone**: password store, wallet keys, sensitive volume and heir policy. The hub releases those keys per action only when its own loop state holds *and* the owner gives a secure-intent gesture on the platform. This brings the lab's terminal-side secure element to the consumer at the scale of a hardware wallet. Two honest limits apply. A laptop's built-in keyboard cannot be put inline, so a laptop owner gets attested input provenance only with an external keyboard. On the built-in keyboard the loop falls back to OS provenance flags, agent-account separation and wrist binding. And on a kernel-compromised machine, the ordinary desktop outside the vaulted zone is protected only on a best-effort basis. Hub hardware, its cost and its certification are not covered by any source in this research.

CONSUMER COMPONENT	PRODUCT AND PRICE EVIDENCE	LOOP ROLE (PROPOSAL)
Custody carrier 1	Apple Watch Series 11 from $399 (ECG); SE 3 from $249 without ECG[185, 186]; Withings ScanWatch launched at $279 with cleared ECG[187]	Anchored custody, pulse liveness, don-time ECG same-body check
Custody carrier 2 (charging overlap)	Oura Ring 4 from $349 plus $69.99 per year; Galaxy Ring $399.99[188, 189]	Second custody carrier so charging never breaks the loop; integration not verified
Identity anchor and secure intent	Built-in Touch ID or ESS face; Magic Keyboard with Touch ID about $179 at a reseller[190]	Anchoring, identity refresh, confirmation gesture
Ambient presence	Aqara FP2 mmWave sensor about $58[142]	Seat presence and second-person trigger; can only lower trust
Optional custody seal	No security-grade patch exists; OTC CGM price anchors: Dexcom Stelo $99 for two 15-day sensors or $89 per month; Abbott Lingo $49 per 14-day sensor[191, 192]	High-assurance sessions only; custom hardware needed
EMG input binding	Meta Neural Band only in a $799 bundle with glasses; no PC pairing found[101]	Not available for v0
Vault hub	Custom; no price evidence	Enforcement point, inline input path, vault keys

The add-on sensor kit of watch, Touch ID keyboard and radar sensor totals about **$640**. Adding a ring brings it to about **$1,000 plus a subscription**. A maximal version built on an Apple Watch Ultra 3 (from $799) runs about **$1,050–$1,400**[185]. Many buyers already own the laptop and the watch, so Vault's marginal hardware is mostly the hub (inference from listed prices).

AI-lab terminal build: the secretless terminal becomes the loop's body

The lab tier keeps the strongest enforcement design already available: a dedicated terminal that never holds model weights or long-lived secrets, talking to a confidential-computing environment that releases keys only on attestation. Anthropic and Pattern Labs described a design in which weights are "stored encrypted, decrypted at the loader, and never released from there", with a keyserver releasing keys only to an attested environment (vendor research)[33]. NVIDIA H100 confidential computing relies on a CPU confidential VM and is attested through NVIDIA's service[193]. Azure confidential VMs with H100 reached general availability in September 2024[194]. Fortanix sells weight release gated on composite CPU and GPU attestation (vendor claim)[195]. What changes is what the terminal attests. It no longer sends "an enrolled face is present". Every one to two seconds it sends signed loop state covering custody, identity refresh, input attribution and contradictions. The lab's verifier appraises that stream, and the keyserver drops the session key the moment it lapses or shows a contradiction (proposal)[34].

LAB COMPONENT	V0 CHOICE (PROPOSAL)	GROUNDING
Terminal	Secretless thin client with tamper-evident chassis; trust engine and attestation key inside a FIPS 140-3 Level 3-class element that zeroizes on intrusion	Level 3 adds tamper detection and response[196]
Identity anchor	Integrated IR camera with frames signed to the secure element and randomized illumination; fingerprint or palm vein plus a FIDO authenticator; USB and virtual cameras blocked	Biometrics only with a physical authenticator[49]; injection evidence[62, 63]
Custody carriers	Enterprise band with fingerprint at don and on-body detection, plus a per-shift adhesive ECG or microneedle seal for the highest tiers	Nymi pattern (vendor)[90]; single-use seal[7]
Input path	Inline IO module in the terminal signs every HID report and renders confirmation UI over the display	ProtectIOn[76]; Not-a-Bot limits[56]
Input binding	Wrist motion per event in v0; EMG as a research track	ZEBRA lessons[18]; EMG evidence[77, 103]
Ambient	Radar occupancy and second-person detection; optional verified co-presence of a second operator	RAND SL5 "supervised access"[68]
Relying party	Confidential VM or keyserver holding a seconds-scale session key; composite CPU and GPU appraisal	Watchdog[34]; TEE.fail forged attestations with a sub-$1,000 interposer[197]
Agents	Agent workloads under distinct identities and mandates; no mandate can grant egress	Windows agent-account model[27]; NIST NCCoE[176]
Irreversible actions	K-of-N biometric-bound quorum plus time delay for export, new egress, quorum or heir changes	Two-person control[198]; RAND SL3[68]

Mapped onto RAND's access benchmarks, per-input attribution exceeds the SL2 requirement of FIDO authentication "for every interaction" or after about 60 seconds[68]. Hardware-rooted enforcement addresses SL4's "secure hardware required for access", and co-presence mode is Vault's contribution to SL5's "supervised access for everyone" inside facility controls[68, 199]. The residual risks remain. The terminal still displays outputs. TEE.fail showed that server attestations can be forged with physical access. NVIDIA GPU reports were not bound to a specific confidential VM, so the verifier must appraise CPU and GPU evidence as one composite[197]. Reference prices exist only for parts: a five-band Nymi starter kit is listed at $1,099 (third-party listing)[200], and a PalmSecure sensor at about $325 from a reseller[201]. Terminal cost, and the cost and schedule of FIPS validation, were not found.

What each build can and cannot promise

THREAT	CONSUMER BUILD	LAB BUILD
Handover after login	Defeated: custody break or new face ends authority at once	Defeated, plus co-presence option
Owner's agent overreaching	Contained in agent workspace; vaulted zone needs owner secure intent	Agent identities cannot reach egress; quorum for agent-initiated export
Injected USB device	Defeated for external peripherals behind the hub; built-in laptop keyboard relies on OS flags	Defeated: only the inline IO module's signed reports count
Virtual-camera deepfake	Defeated on Windows ESS; on macOS the camera can only lower trust	Defeated: signed frames on an integrated camera
Kernel or admin compromise	Vaulted zone protected by hub and secure element; ordinary desktop best effort	Weights never on terminal; session fails closed
Proximity relay	Custody plus secure ranging; Bluetooth presence never counts	Same, inside facility controls
Coercion	Not detected; covert signal, decoy and delays on vault actions	Not detected; covert signal, quorum, delay, quarantined egress

Signal it, shadow it, and make it pay nothing

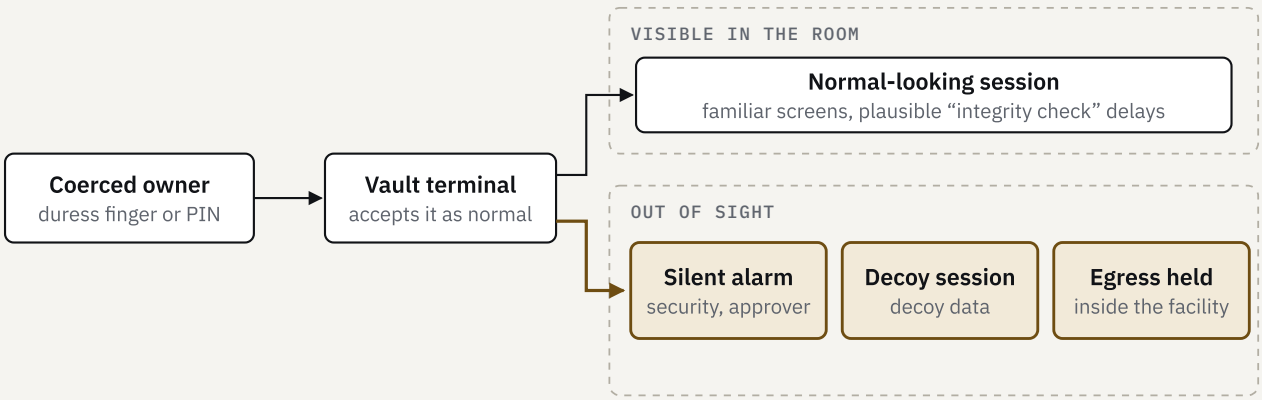
The loop cannot detect coercion, and it raises the stakes

A coerced owner satisfies the first three propositions. The body is alive and attached, it is the enrolled person, and the inputs genuinely come from its hands. No arrangement of channels changes that. The deception-detection literature explains why the fourth proposition cannot be measured instead. The benchmark lab stress classifier reached 93% binary accuracy on 15 mostly young male subjects wearing a chest strap[35]. Field performance drops sharply: one smartwatch study saw F1 fall from 0.84 to 0.71[202], and in a government in-the-wild program no team met its targets[36]. No dataset covers real coercion. Base rates finish the argument. Fifty operators opening four sessions a day produce about 73,000 sessions a year. With genuine duress once a decade, a detector with 95% sensitivity and a 1% false-positive rate raises about 730 false alarms a year (this research's own arithmetic, framed by the National Academies' polygraph analysis and Axelsson's base-rate fallacy)[78, 203]. For EU employees, inferring stress from biometrics is presumptively prohibited when the purpose is protecting property. The Commission's guidelines give "measuring anxiety based on measured stress levels" as their own prohibited example[37].

The loop also raises the stakes. A computer that only one living body can operate makes that body the target, and attackers increasingly go after families: CertiK reports family members of crypto victims increasingly targeted in 2026[69]. The loop does remove two incentives (inference). Liveness and per-event input binding mean that an incapacitated or dead owner, or a stolen device, is worthless to an attacker. They do nothing against a conscious owner held at gunpoint.

What the loop adds: more covert signals and a shadow mode

The loop's channels give the owner more ways to signal duress covertly than a login screen does (proposal). A **duress finger** enrolled on Touch ID or ESS fingerprint behaves normally while raising the alarm, as access-control products already offer[204]. A **duress PIN** opens a working decoy, as Coldcard's trick PINs open a decoy wallet where "nothing special is shown on the screen"[39]. A **duress gesture** could be read by an EMG band, such as a specific muscle pattern with little visible movement. That last idea is attractive because an observer cannot see it, but it is untested: no source evaluates covert EMG signalling or its false-trigger rate under stress, and the FTC's concern that victims "fumble" emergency PINs applies with full force[205]. Whatever the surface, the loop must answer with an **indistinguishable shadow state**. Clark and Hengartner set the formal requirement: the duress credential must be "indistinguishable" by the attacker, any lockout must be "invariant to the type of the passwords being entered", and a visible lockout invites "threatening retaliation"[38]. Decoys must be internally consistent, since operating systems once leaked evidence of TrueCrypt hidden volumes[206]. GrapheneOS's irreversible duress wipe is the wrong default for an owner held by an attacker who expects success[207].



Signal explicitly, degrade silently (proposal). An explicit duress signal never produces a visible lockout or wipe, because those can get a coerced person hurt. The session looks normal while security contacts or the remote approver are alerted, a decoy environment takes over (a decoy wallet on a consumer machine, watermarked canary weights at a lab), and all egress is quarantined inside the facility. A watching attacker can't tell that the signal was given[38, 39, 205].

Make coercion unprofitable on both tiers

The strongest lesson from vault locks, pharmacies and self-custody is to **remove the payoff rather than detect the threat**. Toronto police reported an 82% drop in pharmacy robberies in 2024 after time-delay safes[208]. British Columbia's standards were reported to have cut pharmacy robberies by 94%, and a Calgary detective said time-delay safes "do not lead to an escalation of violence"[40]. These are uncontrolled before-and-after comparisons and a police assertion, not studies. The US Air Force's two-person concept ensures "a lone individual cannot perform an incorrect act or unauthorized procedure"[198]. Bitcoin covenant "vaults" add a window in which a recovery key can claw back a theft[209].

TRIGGER	CONSUMER RESPONSE (PROPOSAL)	LAB RESPONSE (PROPOSAL)	GROUNDING
Any irreversible action, always	Large transfers and heir or policy changes need a delay plus a co-signer (custodian, family member, security firm); the delay is publicized so "I can't" is credible	Export, new egress and quorum changes need a K-of-N live quorum plus delay	Time delays[40]; two-person control[198]
Explicit duress signal (finger, PIN, gesture)	Shadow state: decoy vault opens, alarms go silently to chosen contacts or a protection firm, real keys stay sealed	Shadow state: canary environment, honeytoken credentials, egress quarantined, silent alarm to security and the remote approver	Indistinguishability[38]; decoy wallets[39]; honeywords[210]
Second face or a hand on the owner plus an out-of-scope action	Friction ladder: longer delay, then remote re-confirmation	Friction ladder ending in a two-person requirement; scope shrinks toward read-only	Base rates[203]
Physiological stress features	Not computed; physiology used only for liveness and binding	Not computed; presumptively prohibited for EU staff	[35, 37]

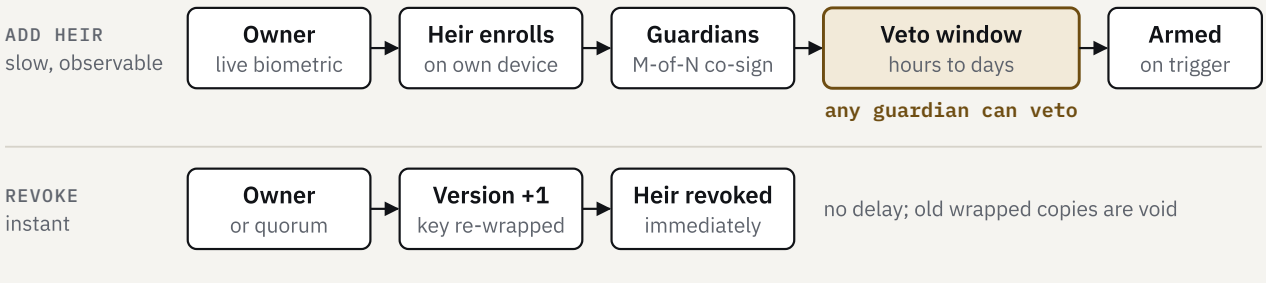
Three practices complete the design (proposal, carried from the duress evidence). Remote approvers and consumer co-signers need their own out-of-band duress channel and should be trained to *appear* to approve while escalating. Coercion often happens at home, so a duress alert should trigger a check on the family. Irish "tiger kidnapping" protocols, which put hostages' safe return first, show why (partially verified parliamentary record)[211]. Any switch the coerced owner can operate alone, such as 1Password's Travel Mode, which removes vaults from a device, is itself a coercion target[212]. Restoring full access after a duress event must therefore need something the owner alone cannot do.

The loop enrolls a new owner, it never lets an heir pass as the old one

Heirs cannot pass the owner's loop, and should not try

An heir is a different body. No arrangement of the loop's channels can make an heir satisfy "this is the enrolled owner" without diluting that proposition for everyone. Succession is therefore a ceremony that *converts* the Vault to a new owner and starts a new loop around a new anchored body. The research found no product binding owner-authorized biometric heirs to hardware, and every shipping inheritance product pairs cryptography with a waiting period and a human or legal attestation[85, 213, 214]. The design below is proposal throughout.

Designation is an owner-signed delegation generated inside the owner's secure element. It binds the heir's device-bound public key to a policy covering activation conditions, scope and expiry. The heir's templates are enrolled on the heir's own hardware and never exported, in line with the EDPB's view that templates on the individual's own device are "in principle compatible" with the GDPR[215]. **Adding an heir** is the most dangerous operation in the system. It needs a live owner loop in normal state, live enrollment of the heir, guardian co-signatures, and a delay of hours to days during which any guardian can veto. **Revocation** is the mirror image: instant for the owner, quorum-only when the owner is unavailable. A coerced owner can be forced to *start* an addition but cannot *complete* it before others intervene. Apple's Stolen Device Protection applies the same asymmetry, requiring biometrics, a one-hour delay and biometrics again before a recovery contact can be changed away from familiar places[3]. A duress signal during "add heir" should make the ceremony appear to succeed while silently freezing it.



Asymmetric delays (proposal). Adding an heir is the most dangerous operation in the system, so it is slow and observable: live owner, live heir on the heir's own hardware, a guardian quorum, then a veto window. Revoking is instant. A coerced owner can start an addition but can't complete it before guardians can step in. A duress signal during the ceremony makes it appear to succeed while silently freezing it[3, 213, 223].

The loop contributes a high-quality owner-liveness signal

The loop does add one thing to succession. It produces the best available evidence that the owner is alive and active, and that evidence can hold the heir path closed. Liana's Bitcoin wallet uses this pattern: the owner periodically "refreshes" coins to keep the heir's relative timelock from opening[216, 217]. In Vault, each normal-state owner epoch could advance a monotonic last-seen counter inside the secure element. The heir path becomes armable only after N days without owner presence *and* trustee attestation of death, incapacity or departure. Storing only the counter, not a presence history, keeps

the succession signal from becoming surveillance (proposal). Absence is not death. Travel, hospitalization and a lost wearable all look like silence, so the timer must be long and the trustee attestation mandatory. Casa's inheritance uses a six-month window the living owner can cancel[85].

Agents need an explicit rule at succession (proposal). When the heir path activates, every mandate the owner signed for an agent should be revoked. An heir who wants agents signs new mandates under the new loop. Otherwise an owner's long-running agent could keep acting for months on behalf of a principal who no longer exists.

Blood, DNA and kinship do not belong in the gate

A kinship or DNA gate fails on every axis. Rapid DNA takes about 90 minutes[6]. DNA transfers to other people's hands in 50% of trials[21]. Half-siblings showed 5–7% false inclusion or exclusion even at 40 STR loci[113]. US employers with 15 or more employees may not request genetic information about an employee or a family member[218], and genetic data is special-category data under the GDPR[219]. Heirs should be authorized by the owner, not identified by biology. A supervised in-person enrollment ceremony provides everything a blood draw would, without the drawbacks.

SUCCESSION ELEMENT	CONSUMER PATTERN	LAB PATTERN
Who succeeds	Named heirs with their own Vault enrollment	Pre-approved successor operators; IANA's Root KSK Ceremony 13 replaced a Crypto Officer[220]
Guardians	Family, lawyer, custodian; Casa-style co-signer[85]	M-of-N custodians; AWS CloudHSM quorum of 2–8 approvers with 10-minute tokens[221]
Activation	Long loop-silence timer plus trustee attestation; Google Inactive Account Manager pattern[222]	Leaver trigger plus custodian attestation
Recovery without biometrics	Sealed paper kit split SLIP-39-style across groups[223]	Sealed, alarmed break-glass accounts in separate safes, validated every 90 days[224]
Highest tier	Optional "no heirs" mode for owners who prefer destruction	"Irrecoverable key policy" at SL5: re-key under a fresh quorum rather than inherit[68]
Agents	All mandates revoked at activation	Agent identities re-issued under successor operators

Verification is defensible on both tiers, inference is the liability

A continuous loop collects far more than a login does: face and fingerprint templates, but also heart signals, muscle activity, typing rhythm, possibly glucose, and observations of everyone else in the room. The legal exposure follows the streams more than the purpose.

Consumer tier: the behavioural and physiological streams carry the heaviest load

For a US consumer product, the two sharpest risks are statutes with private rights of action. Illinois' BIPA requires a written release before collecting biometric identifiers. A 2024 amendment made repeated collection from one person by one method a single violation and allowed electronic signatures[225], and the Seventh Circuit held in April 2026 that the amendment applies retroactively[226]. On-device processing has not been a safe harbor. In *Hazlitt v. Apple*, a case over on-device face clustering in Photos, the court held that BIPA "possession" can apply even where the company does not exclusively control data stored on users' devices[227]. The class was certified on 5 June 2026[228]. Press reports that the Seventh Circuit denied Apple's interlocutory appeal, with exposure estimates of $6.5–32.5 billion (secondary)[229]. An earlier Illinois appellate decision, *Barnett v. Apple*, dismissed claims over biometrics stored only on-device[230]. The case law therefore points both ways, and no court has ruled on on-device *verification*.

Washington's My Health My Data Act is the statute most directly aimed at a loop. "Consumer health data" includes "bodily functions, vital signs, symptoms, or measurements" and biometric data. Biometric data expressly includes "keystroke patterns or rhythms and gait patterns or rhythms that contain identifying information". "Collect" covers anything that will "access, retain, receive, acquire, infer, derive, or otherwise process" such data[42]. Violations are enforceable by the attorney general "as well as through private action"[231], and the first class action was filed against Amazon in 2025[232]. Health *inferences* are in scope, and typing reveals health. Keystroke hold-time features distinguished early Parkinson's patients from controls[233]. A loop's behavioural template drifting with a tremor is therefore itself a health inference. It must never be surfaced, stored or exported (inference). Nevada's similar law is narrower and has no private right of action, and Connecticut now requires consent for consumer health data[234]. Texas's CUBI has no private right, but its attorney general obtained $1.4 billion from Meta and, per legal trade press, $1.375 billion from Google[235, 236].

Federal law adds the FTC. Its 2023 policy statement treats unsubstantiated claims about biometric accuracy as deceptive, and covert collection or failure to assess foreseeable harms as unfair[237]. The Rite Aid order shows the remedy: a five-year ban on facial-recognition surveillance after "thousands of false-positive matches"[238]. The amended Health Breach Notification Rule covers health apps and connected devices outside HIPAA, counts unauthorized disclosure as a breach, and carries penalties up to $51,744 per violation[239]. That matters if Vault ever syncs a heart-rate or glucose stream. The amended COPPA Rule treats children's biometric identifiers, including gait patterns and facial templates, as personal information, with compliance required from 22 April 2026[240].

The EU picture favours the core function. The AI Act's Recital 17 distinguishes biometric *verification*, "including unlocking a device", from remote biometric identification as having "a minor impact on fundamental rights"[241]. An owner's purely personal processing likely falls within the GDPR's household exemption[242]. Whether a device maker is a controller when processing stays local is unsettled. Academics argue that edge processing does not automatically take manufacturers out of scope[243]. Any emotion recognition or biometric categorisation triggers Article 50(3)'s duty to inform, from 2 August 2026[244], and stand-alone high-risk obligations move to 2 December 2027[245].

Two consumer edge cases need product decisions (proposal). The first is bystanders. A loop that watches the room with a camera or radar is processing a spouse's or child's data. Second-person detection should therefore establish only "another person is present", without identifying or retaining anything, and children should receive no camera processing beyond presence. The second is the microneedle channel. A glucose stream is health data in the plainest sense, so the seal should report only "attached and physiologically plausible" to the enforcement point, never readings.

Lab tier: the AI Act bans the tempting feature, and employment law shapes the rest

For lab operators in the EU, the decisive rule is Article 5(1)(f), which prohibits inferring emotions in the workplace except for medical or safety reasons. It has applied since 2 February 2025[246, 247]. The Commission's guidelines list keystrokes, heartbeats and ECG among in-scope signals. They state that "the general monitoring of stress levels at the workplace is not permitted", and they read the safety exception as protecting life and health, "not … property against theft or fraud"[37]. A lab loop that computed stress from its ECG, PPG or EMG streams to detect duress would protect model weights, which are property, and would very likely be prohibited (inference). Verification against a template stored on a device "for the sole purpose of … unlocking a device, or having security access to premises" sits outside the remote-identification ban[37]. Annex III excludes sole-purpose verification from high-risk identification[248]. No guidance addresses *continuous, passive* re-verification, which involves less active participation than a one-time unlock. That is an open question for counsel.

Under the GDPR, employee consent is presumptively invalid[249], so labs need a national legal basis and a necessity case. The Dutch implementing act permits biometrics "necessary for authentication or security purposes". Its explanatory memorandum contrasts a nuclear power plant, where they are likely justified, with a car garage[250]. Enforcement so far has hit time-and-attendance uses: a €725,000 Dutch fine and a €45,000 Belgian one[251, 252]. France's CNIL prefers templates under the person's exclusive control[253]. The EDPB found templates held on the individual's own device "in principle compatible" with the GDPR[215]. A proposed Article 9(2)(l) exception for verification under the data subject's "sole control" was welcomed by the EDPB and EDPS but was still a proposal at last check[254]. In the US, Colorado lets employers require biometric consent to access secure hardware and software, but not for tracking time spent in an application[255]. Washington exempts biometrics collected "in furtherance of a security purpose"[256]. New York bars employers from requiring fingerprints[257]. California requires risk assessments for sensitive personal information from 1 January 2026[258]. At least a dozen states bar employer-required implants[111], and GINA rules out genetic gates[218].

LOOP DATA STREAM	CONSUMER LEGAL HOOK	LAB LEGAL HOOK	DESIGN RESPONSE (PROPOSAL)
Face and fingerprint templates	BIPA release; CUBI; GDPR Art. 9 with household exemption for the owner[225, 242]	GDPR Art. 9 necessity; EDPB on-device position; NY fingerprint rule[215, 257]	Templates only in the owner's secure element; e-signed release at enrollment
Heart signals (ECG, PPG, HRV)	MHMDA "vital signs"; HBNR if disclosed[42, 239]	GDPR health data; AI Act in-scope signal[37]	Used only for liveness and coherence; never stored as a series or exported
Keystroke rhythm, gait, EMG	MHMDA biometric data; COPPA for children[42, 240]	AI Act in-scope signal; Colorado anti-tracking[255]	Per-event binding verdicts only; no behavioural profile leaves the enforcement point
Glucose or interstitial chemistry	Health data under MHMDA and HBNR	GDPR health data	Seal reports only "attached and plausible"
Stress or emotion inference	Art. 50(3) notice in the EU; MHMDA health inference[244]	Presumptively prohibited under Art. 5(1)(f)[37]	Not computed on either tier

LOOP DATA STREAM	CONSUMER LEGAL HOOK	LAB LEGAL HOOK	DESIGN RESPONSE (PROPOSAL)
Bystanders in the room	BIPA, MHMDA and COPPA exposure for non-consenting people	Works-council and notice duties; NY monitoring notice[259]	Presence-only detection with no identification or retention
Presence and attribution logs	FTC unfairness if covert[237]	Function creep toward productivity surveillance[255, 260]	Security events only; succession uses a counter, not a history

Two tiers, one architecture, very different buyers

Consumer tier: high-risk individuals who already pay for security services

Consumer demand is rising, though concentrated. CertiK counted 72 verified crypto wrench attacks in 2025, up 75%, with more than $40.9 million in confirmed losses and Europe accounting for over 40%[41]. Among S&P 500 companies, CEO personal-security perks rose from 35% to 54% prevalence in one year, other named executives' security spending rose from a median of about $10,000 to $32,000, and 2026 proxies increasingly list "digital executive protection"[261]. US border device searches hit a record of about 55,000 in FY2025 (secondary)[262]. Closer to home, an estimated 20% of US adults snooped on someone else's phone in a year[50]. Wearable ownership skews affluent: 31% in households earning $75,000 or more, against 21% overall[143].

Adjacent products show what this group pays for. Casa's custody-and-inheritance plans run from $250 to $7,500 a year, with inheritance included in every individual plan (vendor pricing)[263]. BlackCloak's executive digital-protection plans reportedly cost $3,000–7,000 a year (third-party listing)[264]. Purism sells a $1,999 "Made in USA" phone[265]. The cautionary case is standalone secure hardware. Sirin Labs' $999 Finney phone sold poorly enough that the company laid off about a quarter of its staff in 2019 (secondary)[266]. Free platform features also set a baseline consumers expect without paying: Stolen Device Protection, Identity Check and inactivity reboots[3, 8, 267]. This research infers price anchors of about $10–25 a month for self-serve software on owned hardware, and $2,000–7,500 a year for a high-risk tier with hub, setup, duress protocols and succession. It also infers that the best channels are businesses that already sell security as a recurring service and already hold the succession relationship: crypto custodians, executive-protection firms and family offices. To earn its price, consumer Vault must sell what the free features do not offer: continuous identity rather than custody, coverage of the desktop, duress handling, heir enrollment, and containment of the owner's own agents.

Lab tier: a dozen buyers, rising mandates and an empty intersection

The core lab market is tiny and high-value. Epoch AI counted more than 30 models above 10^25 FLOP from 12 developers as of June 2025[43]. US government pre-deployment testing agreements covered five frontier labs in May 2026 (secondary)[268]. The second ring includes hyperscalers and inference providers that hold partner weights, plus government AI data centers. Mandates create demand without requiring biometrics. California's SB 53 requires frontier developers to publish frameworks for securing unreleased weights, with penalties up to $1 million per violation (law-firm summary)[269]. New York's RAISE Act takes effect on 1 January 2027[270]. The EU code of practice for general-purpose AI asks for "restrictions on device and account sharing" and "periodic personnel integrity testing" (unofficial mirror of the final text)[271]. Lab frameworks still leave the human unverified. Anthropic's ASL-3 protections include "two-party authorization for model weight access" but place sophisticated insider risk out of scope[4]. OpenAI requires hardware-backed security and multi-person approval[5]. The Frontier Model Forum recommends quorum operations and no "static or persistent credentials"[10]. Irregular, the closest comparable company, raised $80 million in September 2025 selling security work to a handful of frontier labs[272].

PRODUCT	TIER	WHAT IT DOES	WHAT IT LACKS FOR A LOOP
Apple SDP, Android Identity Check, Auto Unlock[1, 3, 8]	Consumer	Custody and episodic identity, free	Continuous identity, desktop input provenance, duress, heirs

PRODUCT	TIER	WHAT IT DOES	WHAT IT LACKS FOR A LOOP
Casa, BlackCloak[263, 264]	Consumer	Custody with inheritance; executive digital protection	Any verification loop; natural channel partners
Teleport[273]	Lab	Per-session MFA, hardware keys, dual authorization, moderated sessions	A continuously verified human
CyberArk / Palo Alto Networks[274, 275]	Lab	PAM leader, acquired for about $25 billion	No continuous biometric binding found; has distribution to add it
Okta Identity Threat Protection[276]	Lab	Continuous session-risk re-evaluation	Hardware-attested human
Plurilock[12]	Both	Keystroke and mouse checks "every 3–5 seconds" (vendor claim)	Strong identity; attested input
Nymi Band[90]	Lab-adjacent	Fingerprint band, FIDO2, on-body custody	Presence at the keyboard; input binding
HYPR plus YubiKey Bio[277]	Lab	Fingerprint FIDO authenticator	Continuity
Fortanix Confidential AI[195]	Lab	Attestation-gated weight release (vendor claim)	Any human attestation; likely partner
Windows agent accounts; World AgentKit[27, 169]	Both	Agent identity and human backing	Local per-input attribution

Agent identity is the new hook for both tiers. NIST's NCCoE, the OpenID Foundation and the payment networks are all now asking how to tell a human's action from an agent's[29, 176, 278]. A loop that attributes every input to the owner, the owner's agent or nobody answers that question at the device. Neither the PAM vendors nor the consumer platforms have staked out that position (inference).

One prototype per tier, then the research the loop still needs

One loop, two prototypes, measurable pass marks

Both prototypes should run the same loop software against different enforcement points. The acceptance criteria below are proposals derived from the evidence, not established benchmarks.

WORKSTREAM	CONSUMER PROTOTYPE	LAB PROTOTYPE	PROPOSED ACCEPTANCE CRITERION
Loop core	macOS with Apple Watch and Touch ID keyboard; Windows Copilot+ variant with ESS face; hub prototype inline on desktop peripherals	Secretless terminal streaming signed loop state to a confidential H100 VM holding encrypted mock weights	Authority ends within one epoch (≤2 s) of a contradiction; no plaintext secrets outside the enforcement point
Usability	Ten owners over four weeks, including tattooed, older and adhesive-sensitive users on substitution bundles	Operator pilot over full 40-hour weeks	No more than 5 consecutive false rejects per 40 h[181]; report per-user lockout distributions, not means[127]
Handover and impostor swap	Second person sits down mid-session	Same, plus lent-band and lent-card tests	Swap ends authority on first contradiction; lent carriers go dead on removal
Injection	Rubber Ducky and O.MG-class devices on hub ports; virtual camera on Windows	Virtual-HID driver on host; USB and virtual cameras; IR-frame injection	No injected input reaches the vaulted context; every synthetic event attributed or blocked
Cross-modal forgery	Real-time face swap with attacker wearing an unanchored band	Same, plus radar actuator replay and ZEBRA-style opportunistic mimicry	Agreement among attacker-controlled channels earns no trust; mimicry acceptance well below ZEBRA's 40%[19]
Agents	Desktop agent in an agent workspace attempts vaulted actions	Operations agent attempts weight egress	Zero vaulted or egress actions without owner-present secure intent or quorum
Duress	Duress finger and PIN; decoy vault; silent alert to a contact	Duress signal; canary environment; quarantined egress	Red team iterating across sessions cannot distinguish shadow from normal[38]
Succession	Add-heir ceremony with veto window; loop-silence timer	Successor-operator drill; break-glass drill	A coerced owner can start but not complete an heir addition; revocation is instant
Privacy	No raw biometric or physiological signal leaves the device or hub	Same; EU configuration disables all inference	Reviewed by counsel against BIPA, MHMDA, GDPR and Art. 5(1)(f)

Ten open questions decide whether the loop is buildable

OPEN QUESTION	WHY IT MATTERS	HOW TO RESOLVE
Can EMG or wrist motion attribute individual keystrokes reliably, and resist attack?	P3 has no validated channel; no paper tests EMG-to-keystroke attribution as security	Phase 2 study on open sEMG hardware and Meta's public dataset[279]
Is the ECG-to-facial-pulse delay stable enough to catch forged video?	Rule 8 depends on it; no tolerance found	Bench study with ECG patch and attested camera
How fast does a microneedle or CGM sensor report removal, and can it be fooled in solution?	Decides whether the seal is a seal	Vendor partnership or custom sensor
Can ESS face, or any secure-element path, re-verify continuously through a supported API?	Decides whether passive identity refresh is hardware-rooted on consumer machines	Microsoft engagement; prototype
What does an inline hub cost and how is it certified?	Consumer enforcement and P3 depend on it	Hardware spike; FIPS and Common Criteria quotes
What false-lockout rates do multi-hour sessions with ambient channels produce?	No published data exist	Both pilots
Is on-device continuous verification "collection" under BIPA and MHMDA, and inside the AI Act's verification carve-out?	Sets consent design and EU exposure	Counsel; DPA consultation
Can owners perform covert duress gestures reliably under fear?	No field data for any duress signal[205]	Rehearsal studies; red-team exercises
Will a standard emerge for per-event agent attribution on the OS?	Windows separates agents by account, not by event	Track NCCoE, Web Bot Auth and Windows agent work[174, 176]
Are GPU attestations now bound to a specific confidential VM after TEE.fail?	Prevents borrowed attestations at the lab keyserver[197]	Vendor engagement; composite-appraisal testing

Phase 2 proves the loop end to end and leaves inference out

Phase 2 should deliver the two prototypes above, a hub hardware spike, and three research tracks: per-event EMG and motion input binding, the pulse-transit timing check, and microneedle-seal removal detection. Each should be evaluated against the attacks that broke earlier systems (ZEBRA mimicry, ECG injection, heartbeat-preserving deepfakes), not only against naive impostors. Out of scope for Phase 2: physiological or voice duress inference, DNA or kinship gates for heirs, keys derived from biometrics, mandatory implants or patches, per-key fingerprint keyboards, and SL5 facility controls.

The evidence changes what "only its owner can use it" should mean. Recognizing the owner at the door is a solved problem, and at that point the problem has barely begun. What matters is whether each input is explained by the owner's living body, whether the owner's agents stay inside the authority the owner signed, and whether an owner who is present but not free can be stopped from delivering the payoff alone. A loop built on those three answers turns the founder's two instincts into sound engineering. Every channel stays on in the sense that every channel can stop the machine. And the body's chemistry serves the role it can actually play, as a seal of continuous custody rather than a name.

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Vault · architecture research v2 · October 2026. v2 replaces the gate-based design of v1 (PDF) with an always-on verification loop and adds the consumer tier. This research is not legal advice; regulatory readings are flagged as interpretation where the sources leave them open.